

Better accuracy with fewer qubits: Single-particle basis set optimization for quantum chemistry on quantum computers

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In spite of recent advances, quantum computers are expected to be sufficiently noisy in the coming few years to the extent of limiting atomic and molecular/quantum chemical calculations to relatively small number of orbitals. However, even with reasonable quality single particle basis sets, small active spaces with limited orbitals (fewer qubits) can result in a significant fraction of correlation energy being lost, motivating the design of moderate quality qubit-efficient single particle basis sets with as few orbitals as possible for quantum algorithms. We begin by reoptimizing the existing minimal basis sets using a genetic algorithm-inspired approach in conjunction with aggressive refinement strategies, and generate modified minimal basis sets (which we call MSTO- k G basis, for cardinal number $k = 2$ through 11) for atoms from H through F (excluding He). The minimization is performed at the configuration interaction singles and doubles level of theory. Our open-source codes are available at <https://github.com/subimal/MSTO-kG>. The ground state energies of H through F using our MSTO bases at full configuration interaction (FCI) level of theory yield ground state energies that are comparable or sometimes even better (lower) than those obtained using 6-31G basis sets. In the case of Li, the MSTO bases surpass the performance of the Dunning quadruple zeta (cc-pVQZ) basis sets, all with the same number of qubits as the minimal bases. Thus, we obtain better atomic energies with same number of qubits relative to STO bases, and better/comparable energies with fewer qubits relative to higher quality basis sets. In the case of molecules, H₂ performs poorly; a finding that is consistent with an earlier work in literature. For other molecules, Li₂, C₂, LiH, BeH and BeH₂, the FCI results (except C₂ for which we employ CISD) from our bases are comparable to/outperform those from 6-31G basis. Finally, we compare the resources required between MSTO, 6-31G and cc-pVDZ bases for the case of Li atom picked as a best-case system, and find that MSTO bases yield better energies than the competing basis sets while incurring fewer qubits and two-qubit gates with the NISQ era variational quantum eigensolver approach using the unitary coupled cluster singles and doubles ansatz, the quantum phase estimation (QPE) algorithm applied to complete active space configuration interaction, and the Harrow-Hassidim-Lloyd (HHL) algorithm applied to linearized coupled cluster singles and doubles method. The logical T -gate counts are also found to be considerably lower for QPE and HHL respectively. Overall, our work paves way for more accurate yet less qubit-hungry quantum chemical calculations using near-term quantum computers.

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I. INTRODUCTION

Atomic and molecular physics/quantum chemistry calculations on quantum computers has been a fast-growing field, owing to the promise of speed-up from quantum algorithms such as quantum phase estimation (QPE) [1–3] and more recently the Harrow-Hasidim-Lloyd (HHL) [4–7] algorithms. Long-term implications from such a quantum advantage could range from finding new materials for several real-world applications [8, 9] to drug discovery [10, 11]. In the NISQ era, the variational quantum eigensolver (VQE) algorithm [12] has been the workhorse for quantum chemical calculations on noisy quantum computers.

All of these algorithms have been extended in their ability and scope in several directions. In the case of VQE, the modifications typically occur in the ansatz choice (for example, see Ref. [13] that introduced hardware efficient ansätze, and Ref. [14], which introduced adaptive ansätze via their ADAPT-VQE algorithm) or the optimizer module (for instance, see the recent work by authors in Ref. [15]), to improve its performance in some directions with their corresponding trade-offs. Works in literature also focus on resource reduction techniques including finding equivalent unitaries but with cheaper decomposition and tapering off qubits with little loss in results, which enable executing the algorithm on current noisy hardware and yield fairly accurate results [16–18]. Of late, the problem of executing the QPE algorithm on quantum computers has also been receiving attention owing to the tremendous advances on the quantum hardware and quantum error correction fronts [19]. Several variants of QPE exist that reduce resource requirements (for example, see [20–22]). The authors of Ref. [5] propose application of the HHL algorithm to quantum chemistry, and in the same work, variants of the algorithm improve it. A subsequent work in 2026 shows that one may realize an exponential advantage in the HHL-chemistry framework [7]. Both of these fault-tolerant era algorithms, QPE and HHL, have been executed on current quantum hardware, notably the former was executed with quantum error correction partially baked into their workflow on the Quantinuum quantum computer with reasonable error margins given the scale of the resources required [19].

Existing demonstrations of VQE, QPE, and the HHL algorithms for chemistry problems on real quantum computers indicate that despite the rapid advances on the hardware front, practical computations in the near-term are likely to be restricted to relatively small active spaces (limited orbitals, and therefore a small number of qubits), primarily owing to the result fidelity deteriorating exponentially as we go to deeper circuits for a given gate fidelity. A small active space could result from a low quality basis set such as a minimal basis or picking a subset of orbitals from a high quality basis set such as Dunning’s basis for post-Hartree-Fock (HF) calculations. In the former, the quality of ground state energies is not

very good due to the basis size being insufficient. In the latter, a significant fraction of dynamical correlation effects in predicting ground state energies may be lost, and we also need to rely upon careful selection of orbitals to pick the right ones so that a good part of static correlation can be captured. These considerations motivate the search for approaches to improve achievable accuracy with a fixed (and limited) quantum resource budget. While techniques such as tapering that were mentioned in the earlier paragraph lower qubit count, where the goal is then to achieve comparable accuracy with fewer quantum resources, they introduce additional classical overheads. An alternative to obtaining comparable accuracies with fewer qubits could be indirect resource reduction through *classical* pre-processing routines; for example, use of frozen natural orbitals to lower spin orbital and hence qubit count in VQE (see Ref. [23]). A not-so-pursued approach in the context of indirect resource reduction is to optimize the single particle basis itself; if one could improve upon the contraction coefficients and exponents of existing contracted minimal basis sets and also systematically develop new contracted basis sets of higher cardinal numbers ($k > 6$), we naturally could reach better energies with the same number of spin orbitals/ qubits. We would still be limited by the small size of such bases, but we do the best one possibly can via re-optimization and reaching higher cardinal numbers so that we can extract the maximum possible accuracy with fixed number of qubits. This is the primary goal of our work; we develop a genetic algorithm-based approach in conjunction with aggressive refinement strategies to carry out this exercise for atoms Hydrogen (H) through Fluorine (F) excluding Helium. This exclusion is implied in the remainder of the manuscript. We now expand upon some of the points of this paragraph in the subsequent ones.

We developed our own memetic algorithm to carry out the extension of basis sets to higher cardinal numbers. It is worth emphasizing at this point that although our heuristic algorithms can scale poorly with system size, basis set generation is a one time task, and therefore worth the computational effort. The possibility of using the same number of qubits with better (lower) energy values motivates our search for optimized basis sets and extension to higher cardinal numbers.

We first describe the memetic algorithm that improves upon the exponents and contraction coefficients of Pople’s minimal bases (contracted) from cardinal numbers (k) 2 through 6, that is, the Slater type orbitals (STO) approximated by k weighted and zero-centered Gaussians. We then extend the basis sets to higher cardinal numbers by using the fact that in their contracted form, the number of spin orbitals do not change with the cardinal number. This has the direct implication of requiring fewer qubits in the quantum computing algorithm of interest while yielding better energies. We pick the minimal basis sets in particular, not only because they possess the aforementioned desirable property,

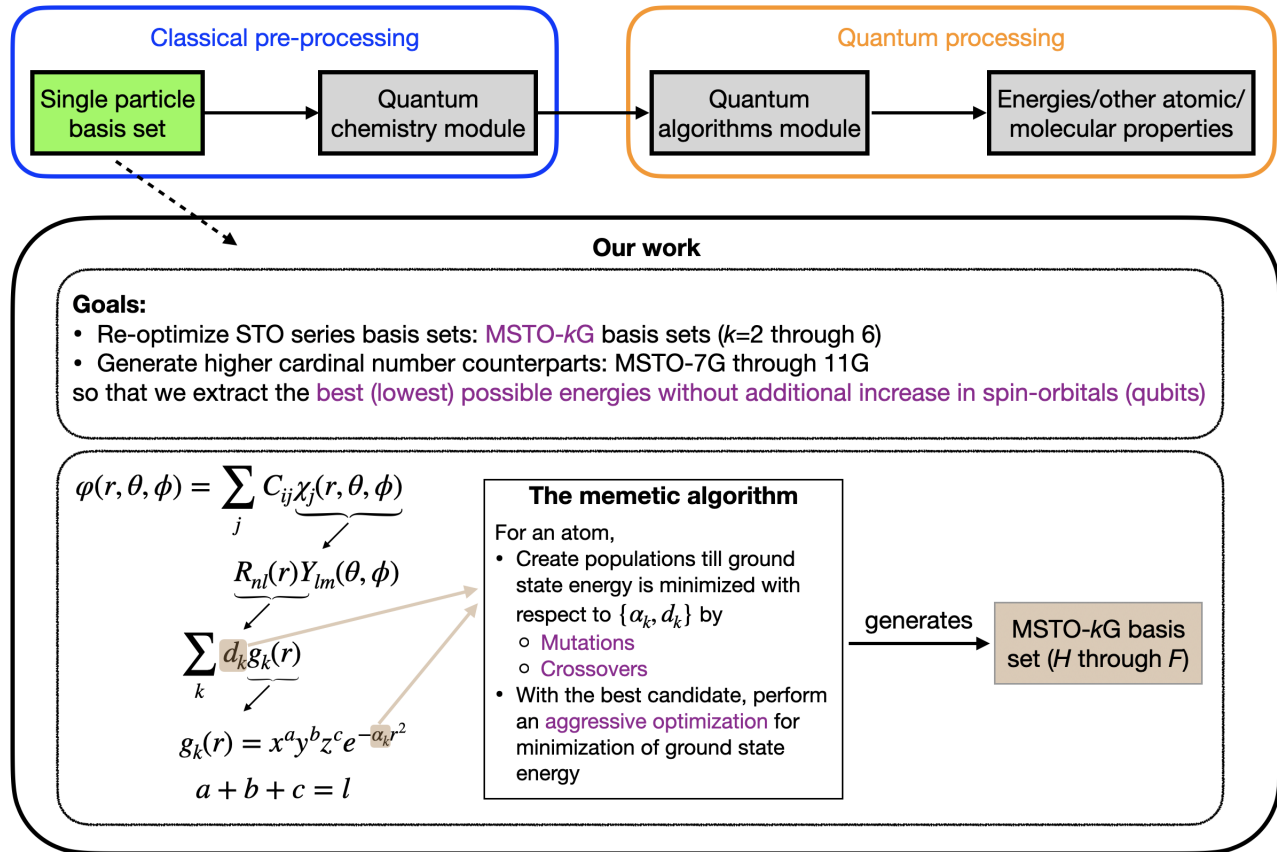


FIG. 1. Schematic of our workflow. A typical quantum chemical calculation in the quantum computing framework involves classical (or quantum-classical hybrid) pre-processing and quantum processing. While there have been many efforts on the processing and the quantum chemical part (Hartree-Fock and integral evaluation) of the pre-processing steps, little work has been done on reducing required quantum resources via basis set optimization. We develop our memetic algorithm to optimize exponents and contraction coefficients of the STO basis sets, extend it to cardinal number 11, and show that the resulting modified STO bases yield lower or comparable energies than the superior split valence basis sets for atoms Hydrogen through Fluorine.

but also because given the current two-qubit gate fidelities on quantum computers, it is sufficient to optimize these bases. We generate the modified STO- k G (MSTO- k G) basis sets for $k = 2$ through 11 for atoms Hydrogen through Fluorine, and benchmark full configuration interaction (FCI) ground state energies for these atoms obtained using our basis sets with FCI values obtained using STO- k G ($k = 2$ through 6), 6-31G, and cc-PVDZ basis sets. A schematic of our workflow and its place in an end-to-end quantum computing workflow for a chemistry problem is presented in Fig. 1.

We note that the STO series bases are of ‘single zeta’ quality and are historically very important, but they have been superseded by modern basis sets. However, they made a comeback in quantum computing, given the field is still nascent, and we expect them to stay for at least a few years given the current status of quantum technology. As an aside, the STO- k G series bases are really not STOs but a linear combination of Gaussian type orbitals (GTOs) that are meant to mimic the behaviour of STOs.

For a review on the work carried out using STOs themselves, interested readers could refer to Ref. [24]. The authors of Ref. [25] provide a good review on basis sets for chemistry in general.

One could always choose a higher quality basis set and work in a small active space, but it can sometimes have the pitfall of yielding very small correlation energies (see Table I of Appendix B), in which case the noisy quantum computer fails to capture the correlation energy accurately. With Pople’s minimal basis sets, the correlation energies within a reasonable active space size are found to be in general larger than those with other basis sets such as Dyal’s or Sapporo’s, or for that matter, even the split-valence basis sets (we will expand upon this in Section IV). Having said that, our memetic algorithm for improving upon exponents and contraction coefficients is general in nature, and can be employed to higher quality basis set families in the future.

The article is organized as follows: Section II briefly discusses minimal basis sets, while in Section III, we ex-

plain our memetic algorithm and its role in optimizing STO- k G basis sets as well as in generating higher cardinal number MSTO- k G basis sets. Section IV presents our results for the ground state energies of atomic systems, Hydrogen through Fluorine, using MSTO- k G basis sets, and their comparison with energies obtained using Pople’s minimal and split valence 6-31G as well as Dunning’s basis sets. This is followed by results for ground state energies and potential energy curves (PECs) for some molecular systems built using the atoms for which we optimized the basis sets. We then discuss in Section V the quantum resources (number of qubits and CX gates) required for VQE, QPE, and HHL algorithms for chemistry calculations, followed by how single particle basis choice affects these resources in relation to ground state energies for the representative example of the Li atom. We then compare our work with other related ones in literature, in Section VI. We finally conclude in Section VII.

II. THEORY AND METHODOLOGY

We can compute the energy of the 1s orbital as follows [26]:

$$E_{1s} = \frac{\int \chi_\mu^* \hat{H} \chi_\mu d\tau}{\int \chi_\mu^* \chi_\mu d\tau}, \quad (1)$$

where $g_{1s}(\alpha) = (2\alpha/\pi)^{3/4} \exp(-\alpha r^2)$ and $\chi_\mu = \sum_{i=1}^k d_i g_{1s}(\alpha_i)$. In the above equation, the integrals are over all space, $d\tau$ being an infinitesimal volume element. Optimized basis sets $\{d_i, \alpha_i\} \forall i$, where d_i refers to the i^{th} contraction coefficient whereas α_i to the i^{th} exponent, can be found either by finding the least squares fit, minimizing the error $\epsilon_{1s} = \int (\chi_{1s}^{STO} - \chi_\mu)^2 d\tau$ (the superscript *STO* refers to the Slater-type orbital) or by minimizing the Rayleigh ratio (Eq. (1)). We chose to minimize the energy of a molecule by gradual perturbation of the atomic basis exponents and contraction coefficients. An optimized basis therefore corresponds to a lower value of the Hartree-Fock (HF) and full configuration interaction (FCI) energies of the molecule of interest for a given k . Pople’s basis sets are available for $2 \leq k \leq 6$. Further, we also obtain optimized basis sets with cardinal numbers up to $k = 11$ providing energy values at least as good as the preceding k .

The STO- k G basis sets were constructed by Hehre *et al* [27] using a linear combination of atomic orbitals self-consistent-field (LCAO SCF) method for a minimal basis set of STOs for atoms not heavier than Fluorine. The representation of the STOs were in terms of GTOs - a sum of k zero-centered Gaussians with a scaling parameter that accounted for the screening effect of each orbital. The parameters of the Gaussian basis sets, $(\{d_k, \alpha_k\})$, were determined by a least square fit of the GTOs to the

corresponding STOs for each principal quantum number. The volume element for the 1s orbital in the least square fit results in using a weight factor $w(r)$ proportional to r^2 for estimating $\{d_i, \alpha_i\}$. As noted by Shavitt in Ref. [28], such a choice of $w(r)$ is appropriate for a three-dimensional integration. It leads to a better fit far away from the origin at the expense of accuracy in the near-nuclear region.

III. OPTIMIZING BASIS SETS WITH A MEMETIC ALGORITHM

For the case of the hydrogen atom, it is straightforward to compute the expression for the ground state energy by computing the integrals in Eq. (1) as

$$E_{1s} = \left[\sum_{i=1}^k \sum_{j=1}^k \pi d_i d_j \left(\frac{4\alpha_i \alpha_j}{\pi^2} \right)^{\frac{3}{4}} \frac{1}{s_{i,j}} \times \left(3 \sqrt{\frac{\pi \alpha_i^2}{s_{i,j}}} - 3 \sqrt{\frac{\pi \alpha_j^2}{s_{i,j}}} - 2 \right) \right] \left[\sum_{i=1}^k \sum_{j=1}^k d_i d_j \left(\frac{4\alpha_i \alpha_j}{s_{i,j}^2} \right)^{\frac{3}{4}} \right]^{-1} \quad (2)$$

where $s_{i,j} = \alpha_i + \alpha_j$. This expression should be minimized with respect to the parameters $\{d_i, \alpha_i\}$, $i = 1, 2, \dots, k$. While using the above expression for minimization of energy, it is to be noted that the contraction coefficients have to be rescaled to normalize the orbital wave functions. From the above expression, it is easy to see that $\alpha_1 = 8/(9\pi)$ for $k = 1$ (for example, see Ref. [26]) leaving d_1 constrained only by the normalization condition on the wavefunction for the 1s orbital in the STO- k G basis. Eq. (2) clearly would not yield simple closed form expressions for atomic basis sets with higher orbitals included. We have therefore resorted to numerical optimization of the ground state energy of the atoms of interest. We next describe our metaheuristic approach to generate the MSTO- k G ($2 \leq k \leq 11$) basis.

We use a memetic algorithm for optimizing the basis sets. A genetic algorithm (GA)-based approach [29, 30] (Algorithm 1) involving mutations (Algorithm 2) and crossovers (Algorithm 3) of candidate basis sets for a given atom in a minimal basis set, and selection of the best candidates (where the objective function is the ground state electronic energy, computed using the Configuration Interaction Singles and Doubles (CISD) method) heuristically optimizes the basis set parameters, often reducing the ground state energy significantly. This is followed by an aggressive gradient-free search (or hill climbing optimization, for example [31, 32]) for a lower ground state energy in the space of k exponents and k contraction coefficients (Algorithm 4) in the modified basis set of STO- k G (MSTO- k G). The hybrid approach combines a global exploration by GA and a local exploration via the gradient-free aggressive search. The

adopted method aims to combine accuracy with computational efficiency.

For a given element, we start with the STO- k G basis sets for $2 \leq k \leq 6$ and apply the memetic algorithm to refine the set for each k . For $k > 6$ we use the MSTO- $(k-1)$ G basis set and add an additional exponent and contraction coefficient (initialized to arbitrary numbers, for example, 1.0 for both the quantities) as a starting point for an MSTO- k G basis. This transfer optimization (see [33] for example) leads to a faster convergence for the higher k values. The tolerance of energy was chosen as 10^{-10} Ha. It is worth noting that the metaheuristic nature of the algorithm ensures convergence of the energy with the chosen tolerance only within the population of the candidates and maximum iterations fixed for the search and therefore may not be the analytical minimum. Multiple passes of the memetic algorithm are used to reach as close as possible to the optimal value. Further, for the memetic algorithm part, $N = 10$ candidate basis sets were prepared from the bootstrapped basis which can limit the search space for large k . Although the value chosen for N can reduce the search space, a higher value may make the algorithm very resource intensive. The trade-off has to be decided by the user generating the basis sets based on the available hardware resources. Nevertheless, we have achieved ground state energies that are very close to that obtained from the 6-31G basis sets with better results in most cases.

We use these basis sets and compute the ground state energy of a molecule from combinations of the standard minimal basis set and atom-optimized MSTO- k G bases, and compare with the 6-31G basis. Except for systems with a hydrogen atom, the energy landscape of the studied molecules outperforms or is comparable to that of 6-31G.

The exponents and contraction coefficients for the MSTO- k G basis sets for k from 2 through 11, and for atoms H through F , are given in Appendix A.

Algorithm 1 Genetic Algorithm Part of the Hybrid Memetic Optimization for MSTO- k G Basis

Input: Atom M , Population size N , Max trials T , Tolerance τ

Output: Optimized basis set parameters Θ_{best}

```

1:  $Pop \leftarrow \text{Generate\_Initial\_Population}(M, N)$   $\triangleright$  Generate
   candidates by mutations of  $M$ .
2: Sort  $Pop$  by  $E$  (CISD energy) in ascending order
3: for  $t = 1$  to  $T$  do
4:    $f \leftarrow \text{Dynamic\_Random\_Factor}(t)$   $\triangleright$  Factor based on
     current trial
5:    $Muts \leftarrow \text{Mutate}(Pop, f)$   $\triangleright$  Interpolative mutation
6:    $Pop \leftarrow \text{Keep\_Best}(Pop \cup Muts, N)$   $\triangleright$  Elitism
7:   for  $j = 1$  to  $N_{co}$  do  $\triangleright N_{co}$  is number of crossovers
8:      $Cross \leftarrow \text{Cross\_Overs}(Pop)$   $\triangleright$  Discrete parameter
       swapping  $d_i \leftrightarrow d_j$  or  $\alpha_i \leftrightarrow \alpha_j$ 
9:      $Pop \leftarrow \text{Keep\_Best}(Pop \cup Cross, N)$ 
10:   end for
11:    $E_{best} \leftarrow Pop[0].E$ ,  $E_{worst} \leftarrow Pop[N-1].E$ 
12:   if  $|E_{best} - E_{worst}| < \tau$  then  $\triangleright$  Check for convergence
13:     break
14:   end if
15: end for
16: return  $Pop[0]$   $\triangleright$  The best candidate after an iteration of
    mutations and crossovers is the new solution.
```

Algorithm 2 Interpolative Mutation (Mutate)

Require: Population Pop , Dynamic random factor f

Ensure: List of mutated candidates $MList$

```

1:  $N \leftarrow \text{length}(Pop)$ 
2:  $MList \leftarrow [Pop[0]]$   $\triangleright$  Retain best candidate
    $\triangleright$  Generate new candidates by tweaking parameters from
   candidate pairs to intermediate values.
3: for  $i = 0$  to  $N - 2$  do
4:   for  $j = i + 1$  to  $N - 1$  do
5:      $\Theta_i, \Theta_j \leftarrow \text{Parameters of } Pop[i], Pop[j]$ 
6:      $\Delta\Theta \leftarrow (\Theta_j - \Theta_i) \times f$ 
7:      $\Theta_{new} \leftarrow \Theta_i + \Delta\Theta$ 
8:     try  $\triangleright$  Evaluate  $E$  of the new candidate
9:        $E_{new} \leftarrow \text{getEnergies}(\text{Build}(\Theta_{new}))$   $\triangleright$ 
       Add this candidate to the list if it is better than the best
       candidate in the existing list. Keep it sorted by  $E$ .
10:      if  $E_{new} < E$  of  $(MList[0])$  then
11:         $MList.append([E_{new}, \Theta_{new}])$ 
12:        Sort  $MList$  and truncate to size  $N$ 
13:      end if
14:    end try
15:   end for
16: end for
17: return  $MList$ 
```

Algorithm 3 Discrete Parameter Swapping (Cross_Overs)

Require: Population Pop
Ensure: Crossover offspring $CList$

```

1:  $N \leftarrow \text{length}(Pop)$ 
2:  $Idx \leftarrow \text{Shuffle}(\text{indices of } Pop)$ 
3:  $N_{limit} \leftarrow \lfloor N \times 0.75 \rfloor$   $\triangleright$  75% crossover rate
4: for  $k = 1$  to  $N_{limit}$  do
5:    $p1, p2 \leftarrow Idx[0], Idx[1]$   $\triangleright$  Select two parents
6:    $\Theta_1, \Theta_2 \leftarrow \text{Parameters of } Pop[p1], Pop[p2]$ 
7:    $T \leftarrow \text{Random\_Binary\_Template}(\text{shape of } \Theta)$ 
8:    $\Theta_{off} \leftarrow (T \times \Theta_1) + ((1 - T) \times \Theta_2)$   $\triangleright$  Exchange a few
     of the parameters to create 2 offsprings.
9:   try
10:     $E_{off} \leftarrow \text{getEnergies}(\text{Build}(\Theta_{off}))$ 
11:     $CList.append([E_{off}, \Theta_{off}])$ 
12:   end try
13: end for
14: return  $CList$ 

```

Algorithm 4 Parallel Aggressive Refinement

Require: Candidate C , Sample size μ , Precision factors F , Cores n
Ensure: Refined candidate C_{final}

```

1:  $E_{best} \leftarrow \text{getEnergies}(C)$   $\triangleright$  Mutate the best candidate by
   iteratively decreasing perturbations (e.g., 0.1, 0.01...)
2: for  $f \in F$  do
3:    $Improved \leftarrow \text{True}$ 
4:   while  $Improved$  and  $iters < itermax$  do
5:      $Improved \leftarrow \text{False}$   $\triangleright$ 
     The mutations are done in parallel saving the candidate
     upon improving GS energy. The same perturbation scale
     is used as long as improvements are obtained.
6:      $Jobs \leftarrow \text{Parallel\_Sampling}(C, f, \mu, n)$   $\triangleright$  Joblib
       backend
7:      $C_{new} \leftarrow \text{argmin}_E(Jobs)$ 
8:     if  $E(C_{new}) < E_{best}$  then
9:        $C \leftarrow C_{new}$ ,  $E_{best} \leftarrow E(C_{new})$ 
10:       $Improved \leftarrow \text{True}$ 
11:      Save  $C$  to file
12:     end if
13:      $iters \leftarrow iters + 1$ 
14:   end while
15: end for
16: return  $C$ 

```

IV. RESULTS

A. Atomic ground state energies with our basis sets

Figures 2 to 5 present HF and FCI results using:

1. the minimal basis sets (k of 2 through 6),
2. our MSTO- k G (k of 2 through 11),
3. the split-valence 6-31G, and

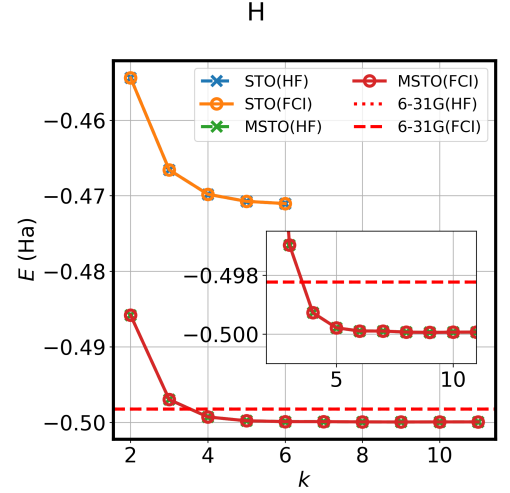


FIG. 2. HF and FCI ground state energies of the hydrogen atom from MSTO- k G basis, compared with minimal basis sets and split valence basis sets (6-31G).

4. cc-pVXZ (X=D, T, Q in Fig. 3),

with the accompanying data in Tables II through IX of Appendix B. The tables in the Appendix also provides energies from the optimized STO- k G ($k = 3, 6$) basis sets optimized by generalized simulated annealing [34], STO- k G ($3 \leq k \leq 6$) basis sets by Hehere *et al* and the exact values for STOs (the last two are taken from [27]). For the case of Li, since our basis performed comparable to Dunning's cc-pVQZ basis in predicting ground state energies, we have added the results from the basis too (Fig. 3(a)). We exclude He since by construction, noble gas atoms have no virtuals and hence no correlation effects in the minimal basis. Our MSTO- k G basis inherits the same trait. The PySCF package [35, 36] was used for all the computations.

Fig. 2 shows that for Hydrogen, the ground state energy using the MSTO- k G basis practically reaches the theoretical value of -0.5 Ha. The case of the H atom is exactly solvable and this exercise helps in assessing the correctness of our code. At $k = 4$, our MSTO- k G basis surpasses the ground state energy from the 6-31G basis. The correlation energy was found to be consistent with zero (within a numerical error of about 10^{-16} Ha) in STO and MSTO bases, with the reason being the absence of electron-electron interaction in the Hamiltonian of the one-electron atom.

For Lithium, Fig. 3(a) shows that the ground state energy from the MSTO- k G basis surpasses even that obtained with the cc-pVQZ basis, and additionally also resolves the correlation energy in contrast to the STO, 6-31G, and the cc-pVQZ bases (Fig. 3(b)). In fact, our result using the MSTO-11G basis (-7.452345 Ha) is within 20 mHa of the benchmark ground state energy results from Table I of [37] (-7.478060 Ha, the Hylleraas ∞ basis quoted in Table III of Appendix B).

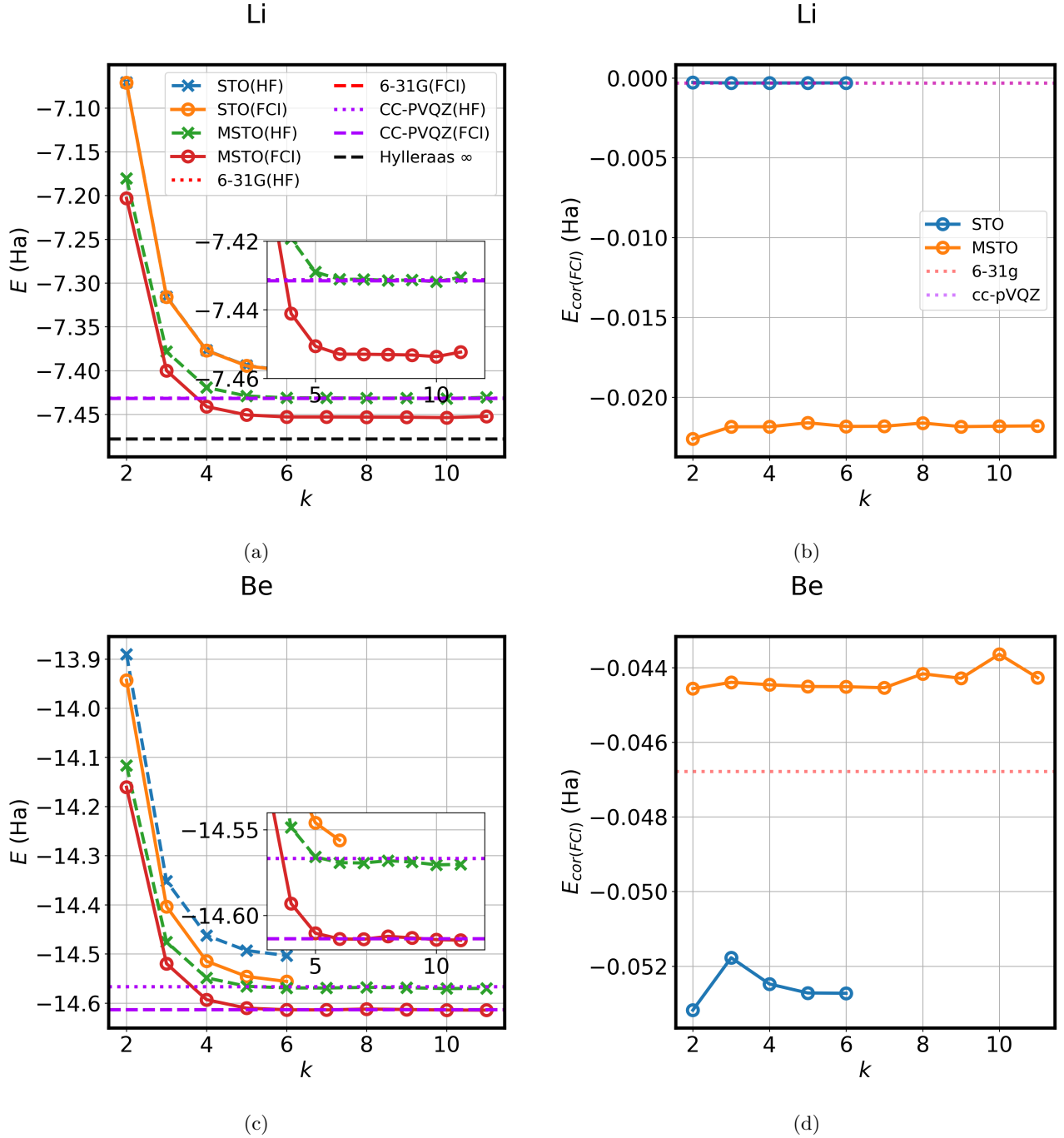


FIG. 3. HF and FCI ground state energies of (a) Li and (c) Be with our MSTO- k G basis, compared with minimal basis sets and split valence basis sets (6-31G). The energies for 6-31G and cc-pVQZ cannot be resolved at the scale of the figure. The FCI correlation energies captured for (b) Li and (d) Be for MSTO- k G basis compared against the minimal STO- k G basis sets show a significant improvement for the new basis along with an apparent saturation beyond $k = 6$ (see Tables III and IV).

For Beryllium, Fig. 3 (c) shows that the MSTO- k G basis yield ground state energies nearly matching the split-valence basis and cc-pVXZ (X=D, T, Q) bases. The MSTO-11G basis performs better than the 6-31G, cc-pVTZ and cc-pVQZ bases by about 1 mHa but falls short of the value from cc-pVDZ basis by about 3 mHa (see Table IV of Appendix B).

Fig. 4(a) and (b) present our results for Boron and Carbon, respectively. We see that the MSTO bases consistently outperform STO ones and predict ground state energies comparable to those from 6-31G basis, falling short of the higher quality basis sets by about 10 and 20 mHa, respectively (see Tables V and VI of Appendix B). We also observe that for these two systems and for

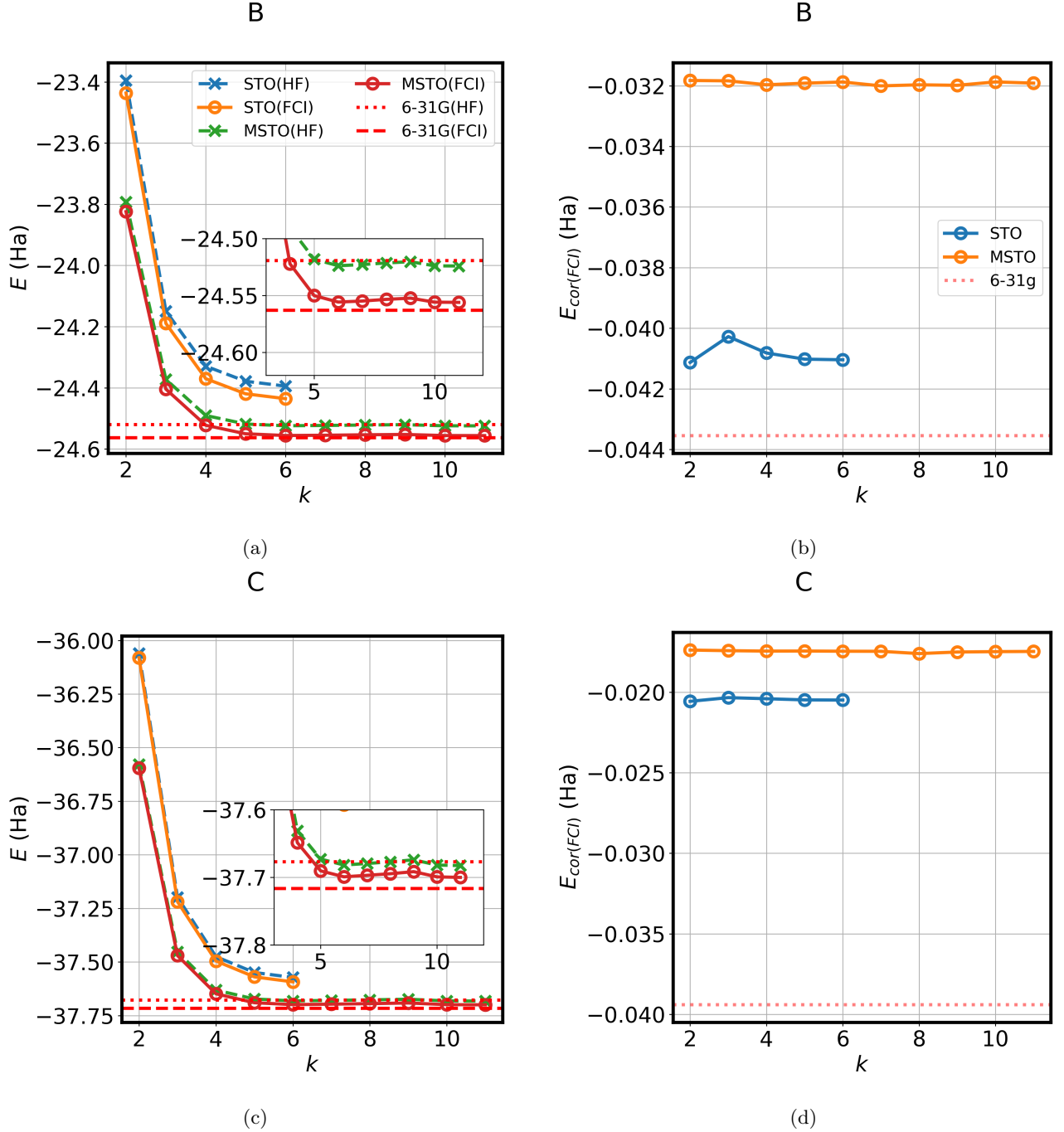


FIG. 4. HF and FCI ground state energies of (a) B and (c) C with our MSTO- k G basis, compared with minimal basis sets and split valence basis sets (6-31G). The FCI correlation energies captured for (b) B and (d) C for MSTO- k G basis compared against the minimal STO- k G basis sets show a significant improvement for the new basis along with an apparent saturation beyond $k = 6$ (see Tables V and VI).

Beryllium, while the quality of the energies are better than the STO bases, the correlation energies obtained using them are smaller than those from the STO bases, thus making it easier for a quantum computer to capture the correlation energies in the MSTO- k G basis. However, we find that the correlation energies from higher quality basis sets in an active space of 10 spin orbitals are much

smaller (see Table I of Appendix B). For example, with the 6-31G basis, when we work with the first 10 spin orbitals, we recover only 55% of the correlation energy for B and only 33% for C. If we instead check this behaviour for the larger cc-pVDZ basis, we only recover 29% and 15% of the correlation energies for B and C respectively. We note that this is only for illustration; in practice, one

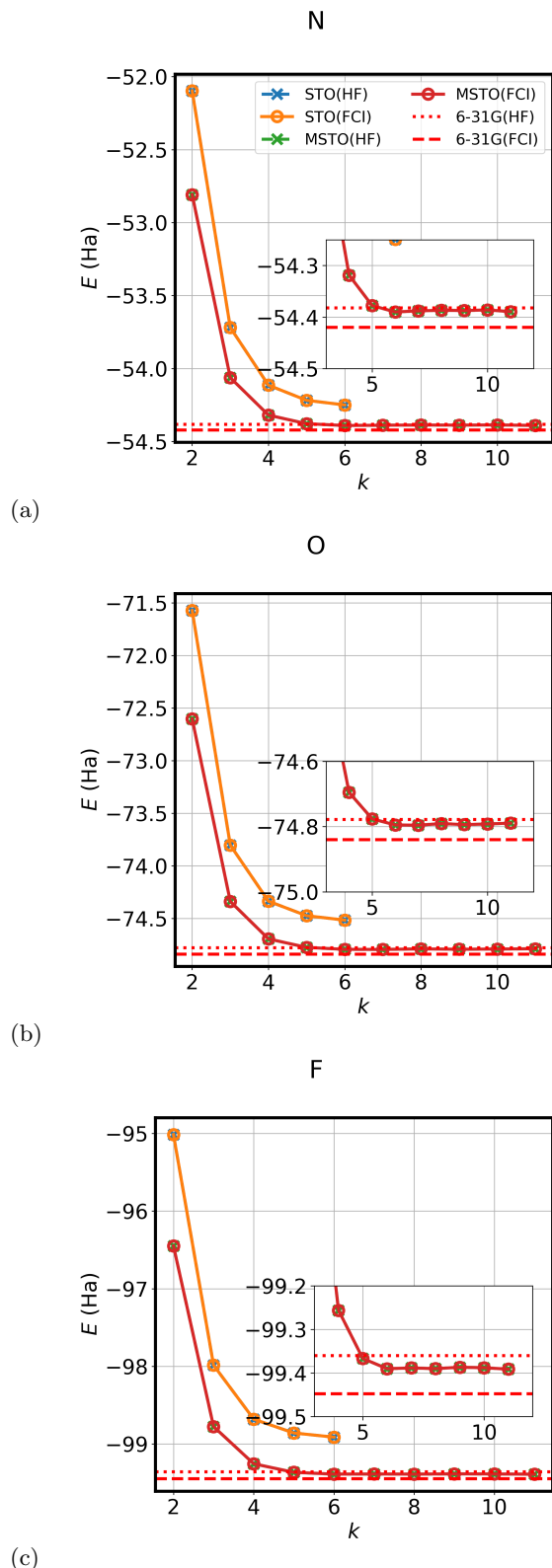


FIG. 5. HF and FCI ground state energies of (a) Nitrogen, (b) Oxygen, and (c) Fluorine from MSTO- k G basis, compared with minimal basis sets and split valence basis sets (6-31G). The FCI correlation energies are near-zero (see Tables VII, VIII and IX).

invokes chemical intuition to pick the relevant orbitals for an active space, but it is not always feasible to recover the most of correlation energy with the right choice of orbitals as one would still miss dynamical correlation with a small active space. Thus, the MSTO bases offer a reasonable trade-off between the quality of results and the size of the correlation energies.

We do not expect further optimization to lead to substantial improvements in the ground state energies. We expect that we have likely extracted the most one could from the minimal basis sets, which require a limited space of orbitals.

While we observe the superior performance of our MSTO basis sets over the STO ones in predicting HF energies for N, O, and F, it is worth noting that our bases inherit the same limitation as the STO ones in that for these 3 atoms, they do not have sufficient number of virtuals to generate any noticeable contribution to correlation energies. Improving this limitation by possibly adding diffuse and/or higher angular momenta functions is beyond the scope of the current work, and we defer it to a future study.

B. Molecular ground state energies with our basis sets

We pick the simple H_2 , Li_2 , and C_2 molecules for computing their potential energy curves (PECs) with the STO-6G, MSTO-6G, MSTO-11G, and the 6-31G basis sets. The results presented in Fig. 6(a) and (c) show that for the H_2 and C_2 molecules, the MSTO bases, which nearly overlap with one another across the HF and FCI PECs, perform worse relative to STO-6G and 6-31G basis sets in the neighbourhood of their corresponding equilibrium bond lengths. This observation is consistent with that from Andrade *et al* [34]. Furthermore, the equilibrium bond length from MSTO bases itself is slightly longer than the ones predicted by STO-6G and 6-31G bases. However, as we move away to stretched geometries away from the equilibrium bond lengths, the MSTO bases yield lower FCI energies than the other basis sets considered.

For the case of Li_2 (Fig. 6 (b)), we find that the FCI energies obtained using MSTO-6G and MSTO-11G bases consistently outperform the 6-31G results. Just as in the case of the H_2 molecule, the MSTO bases predict slightly longer bond lengths than the minimal and the split-valence basis sets.

We study the performance of STO and MSTO bases as well as hybrid ones (MSTO for one atom and STO for the other) for three heteronuclear diatomic molecules: LiH , BeH (Fig. 7), and BeH_2 (Fig. 8). For the case of LiH , the combination of MSTO-MSTO bases for Li-H consistently predict lower energies than the STO-STO combination and the 6-31G basis, and while MSTO-STO and STO-MSTO combinations fare worse than MSTO-MSTO and 6-31G basis sets, they outperform the STO-

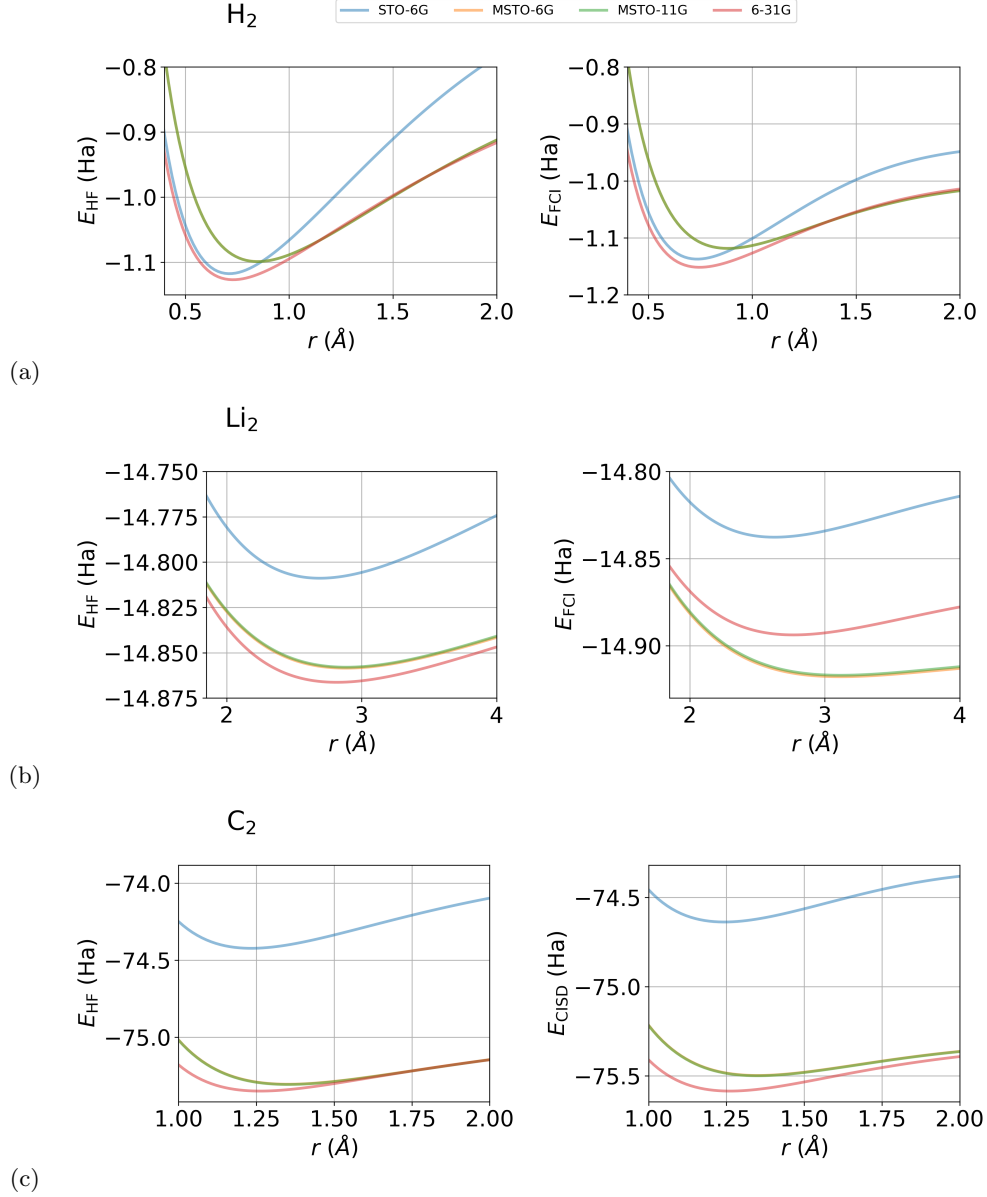


FIG. 6. The HF (left panels), FCI (right panels) PECs for (a) H_2 and (b) Li_2 and CISD (right panel) for (c) C_2 . The PECs for the atom optimized MSTO-6G and MSTO-11G bases are compared with the standard basis sets, namely STO-6G and 6-31G bases. All plots share the same legend. In all the plots the curves for MSTO-6G and MSTO-11G coincide.

STO bases. We also observe that the MSTO-MSTO, 6-31G, and hybrid bases all predict comparable equilibrium bond lengths at the FCI level, all slightly longer than the one predicted using STO-STO basis sets. This observation is consistent with that from an earlier work [34].

V. RESOURCE ESTIMATION IN QUANTUM ALGORITHMS WITH OUR BASIS SETS

We now discuss estimates of the quantum resources (number of CX gates and number of qubits) required for

carrying out a VQE-UCCSD ground state energy calculation. We then perform the same exercise for the fault-tolerant era algorithms, QPE-CASCI and HHL-LCCSD approaches. Carrying out a detailed resource estimate (such as the number of physical qubits and its trade-off with runtime, etc) for these two algorithms is outside the scope of our work, and we defer it for a future study. We restrict ourselves only to estimates of the number of logical qubits and the number of CX gates (and logical T gates) and comment on how choice of single-particle basis can affect the estimates in relation to the predicted quality of energies.

Following estimates for the number of logical qubits

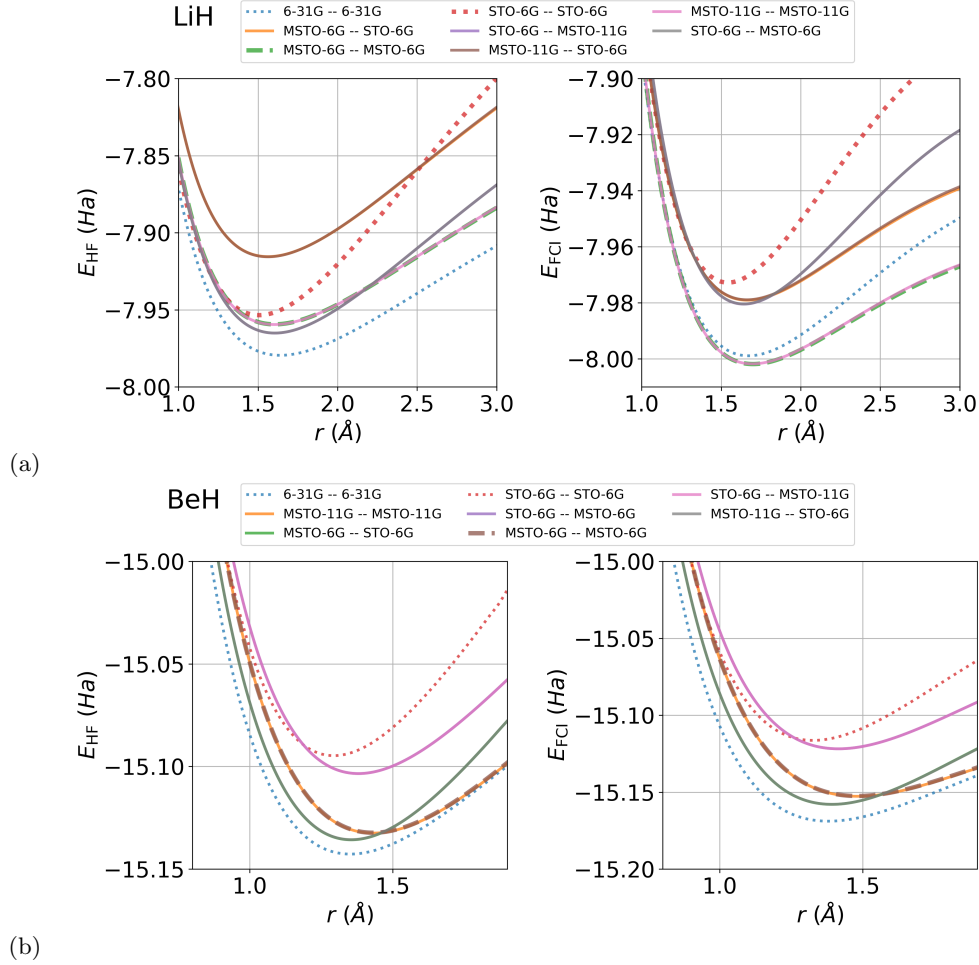


FIG. 7. The HF (left panels) and FCI (right panels) PECs for (a) LiH and (b) BeH. The PECs for the atom optimized MSTO-6G and MSTO-11G bases are compared with the standard basis sets, namely STO-6G and 6-31G bases. The order of basis for the atoms are in the same order as the atoms in the formula. All plots share the same legend. In all the plots the following pairs of curves are coincident: (i) MSTO-6G–STO-6G with MSTO-11G–STO-6G, (ii) MSTO-6G–MSTO-6G with MSTO-11G–MSTO-11G, and (iii) STO-6G–MSTO-6G with STO-6G–MSTO-11G.

and the CX gate count, we pick *Li* as a representative system (which gave the best energies with our MSTO bases) and compare these two quantities against the FCI ground state energies obtained using STO-6G, MSTO-6G, 6-31G, and cc-pVDZ bases. We recall that ideally, we would want a single particle basis set to require relatively fewer qubits and incur fewer CX gates while yielding better (lower) energies.

We note a few points about the scope of this analysis:

1. For our analysis, we only consider the quantum algorithms in their vanilla form. This is for simplicity, as often, reduction in quantum resources that occur in different variants of an algorithm either incurs classical computing overheads that complicate estimating costs (for example, a recently introduced quantum-information inspired ansatz for VQE lowers gate and parameter counts but requires knowing the logarithm of an approximately computed density matrix [38]) or the improved variant

is at the level of showing that such an algorithm exists and for which circuit designs may not be obvious (the Childs-Kothari-Somma improvement over HHL [39]).

2. For the *Li* atom example, we use the $\{RZ, X, SX, CX\}$ basis as the native gate set. Using other choices would change the gate counts but not the overall observed trends.
3. In view of the cost, we do not compute energies using the quantum algorithms themselves, but rather just use the FCI energies in each basis as proxies for the energies that those algorithms would yield. This approximation is reasonable, since the HF contribution for a system, which constitutes almost all of the total energy, varies substantially across the basis sets considered. In contrast, the additional correlation energy contribution recovered with UCCSD, FCI, or LCC would be a small frac-

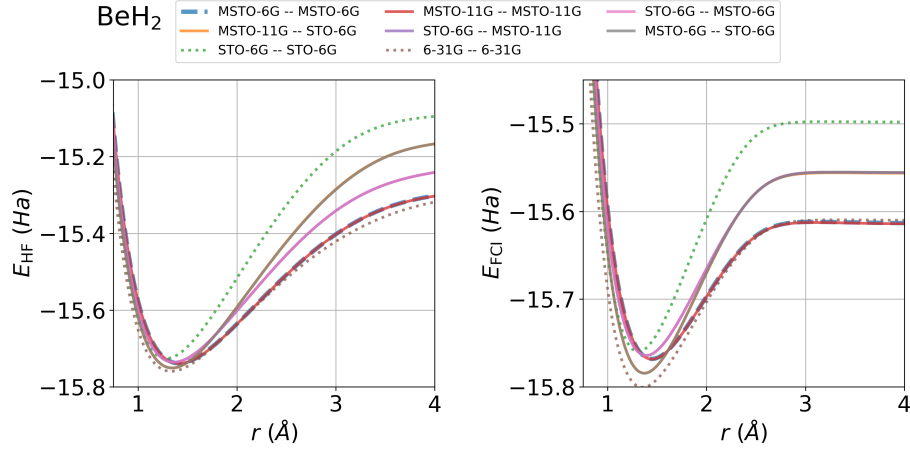


FIG. 8. The HF (left panel) and FCI (right panel) PECs for BeH_2 . The PECs for the atom optimized MSTO-6G and MSTO-11G bases are compared with the standard basis sets, namely STO-6G and 6-31G bases. The order of basis for the atoms are in the same order as the atoms in the chemical formula. Both the plots share the same legend. In all the plots, the following pairs of curves are coincident: (i) MSTO-6G–STO-6G with MSTO-11G–STO-6G, (ii) MSTO-6G–MSTO-6G with MSTO-11G–MSTO-11G, and (iii) STO-6G–MSTO-6G with STO-6G–MSTO-11G.

tion of the HF energy. Thus, for the purposes of our qualitative resource estimates, using the FCI energy as a common reference is reasonable.

A. VQE-UCCSD

The number of qubits in VQE (n) is the number of spin orbitals, n_s . We use the UCCSD ansatz, and hence the number of parameters scale in general as $\sim n^4$. In practice, the number is lesser as we shall see in our numerical example. Upon Trotterizing (first order and first step, chosen for the purposes of simplicity), each of the n^4 terms of the form $e^{\theta_i P_i}$, where θ_i is the i^{th} parameter/amplitude (usually up to a factor of 2 or its integral multiple) and P_i is an n -qubit Pauli word, can be compactly represented using a Pauli gadget that has $\sim 2(n-1)$ CX gates. Thus, we have about $\sim n^5$ two-qubit gates per expectation value evaluation, $\langle \Psi(\vec{\theta}) | H_i | \Psi(\vec{\theta}) \rangle$, per iteration. In the subsequent paragraphs, when we present our results for the number of CX gates, it is the quantity per expectation value evaluation per iteration.

The data in Fig. 9(a) shows that with the MSTO-11G basis, we incur the same number of qubits and same number of CX gates as with the STO basis, but with a substantially lower ground state energy. On the other hand, we require far fewer qubits and CX gates with MSTO relative to 6-31G (about 56% of the qubits and about 13% of the CX gates) or cc-pVDZ (about 36% of the qubits and about 3% of the CX gates) bases but at the same time, we also obtain better energies. We emphasize that Li is the best case scenario, and in other atoms, we do not outperform the Dunning basis sets but arrive at accuracies better than or close to the 6-31G bases. Sub-figure (b) of Fig. 9 shows that we obtain lower energy with the

MSTO-11G basis with far fewer number of variational parameters relative to 6-31G or the cc-pVDZ bases (also see supporting data in Table X of Appendix C).

B. QPE-CASCI

We now move to gate counts and energies from QPE. The number of qubits (n) here is the sum of the number of spin orbitals (n_s) and the number of clock register qubits (n_r) to capture the eigenvalues to within n_r digits of precision in their binary representation. We arrive at the estimate for CX gate count by:

- expanding $U = e^{iHt} = e^{i\sum_{i=1}^l H_i t} \approx \prod_{i=1}^l e^{iH_i t}$ (we assume first order Trotterization with Trotter step set to 1 for simplicity),
- seeing that a Pauli gadget that represents each $e^{iH_i t}$ has $2(n_s - 1)$ CX gates [40] and thus l such gadgets for U yields $2l(n_s - 1)$ CX gates,
- using a result from Ref. [41] that shows that a $(n_s + 1)$ -qubit controlled unitary where $U \in 2^{n_s} \times 2^{n_s}$ has $\approx 2(n_s + 1)$ -qubit Pauli gadgets to arrive at an estimate of $4ln_s$ CX gates per controlled unitary, and finally
- since there are about 2^{n_r} calls to CU in QPE, we obtain a total of $\approx 2^{n_r} 4ln_s$ CX gates.

The QPE algorithm also has an inverse quantum Fourier transform (IQFT) module, which has $\approx n_r(n_r - 1)/2 + 3n_r/2$ CX gates. The latter comes from the fact that there are about $n_r/2$ SWAP gates, and each SWAP gate can be decomposed into a sequence of 3 CX gates.

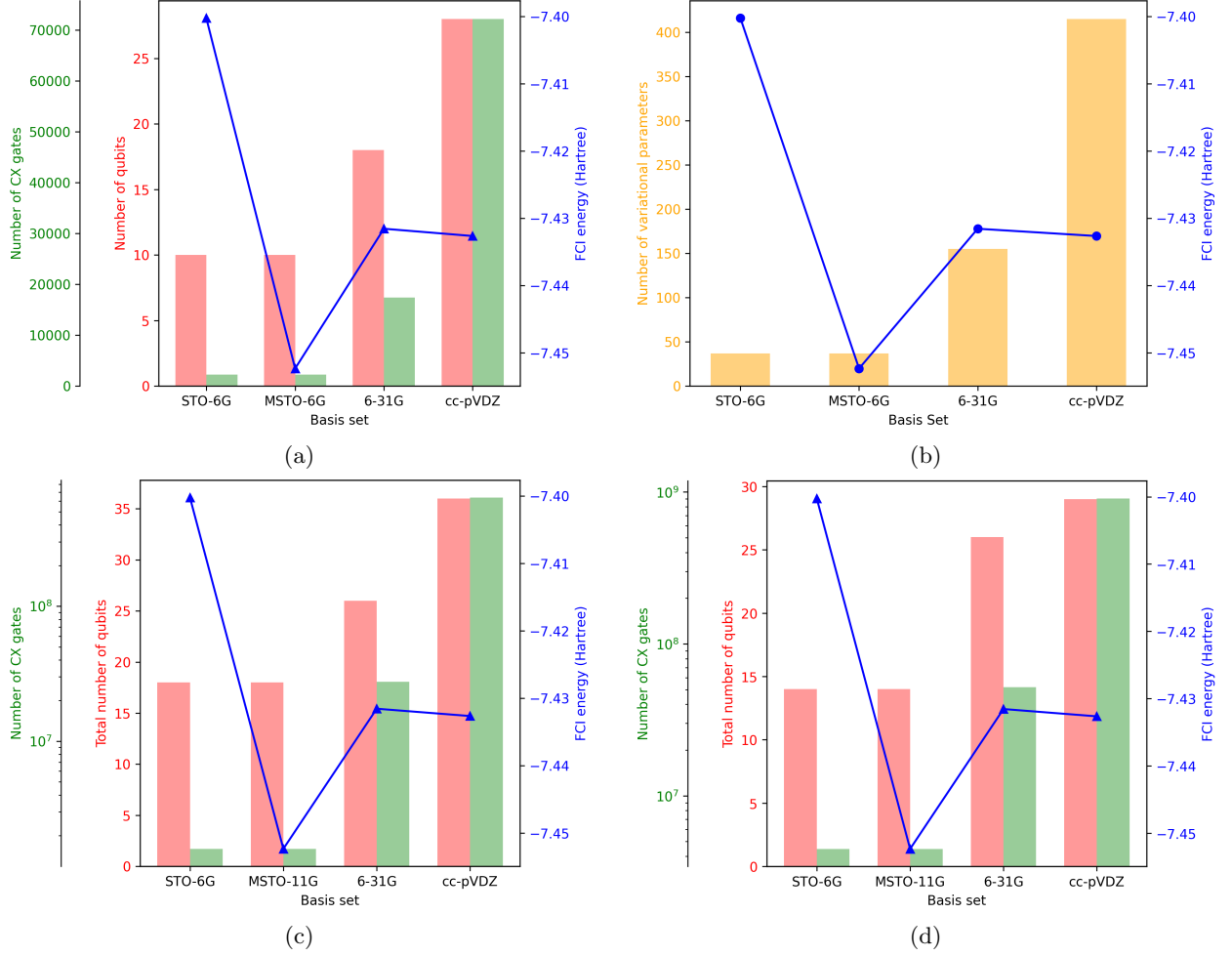


FIG. 9. Comparison of resource requirements across different basis sets, for the Li atom. The top row presents our results for VQE (circuit resources and variational parameters), while the bottom row shows the QPE-CASCI (sub-figure (c)) and the HHL-LCCSD (sub-figure (d)) results (total qubits and CX gates).

Therefore, the CX gate estimate for QPE is $\approx 2^{n_r} 4ln_s + n_r^2/2 + n_r$. We make three remarks at this juncture:

- The number of clock register qubits, n_r , grows very slowly with system size, and hence the gate count at the asymptotic limit is governed primarily by n_s , that is, the number of spin orbitals.
- For the same reason, QPE does not have an exponential number of gates, but only polynomial (in n_s) number of them.
- Typically, for molecular Hamiltonians, $l \sim \mathcal{O}(n_s^4)$, due to the Hamiltonian containing at most pairwise Coulomb interactions. Upon Jordan-Wigner transforming the Hamiltonian, one typically retains this scaling. In practice, the number of terms are fewer than this worst case estimate.
- Although we focus on CX gate count, since all the circuits would be executed at the logical level, the

T gate count is also very important to estimate, as they cannot be realized transversally in popular codes such as the surface code [42] and the Steane's code [43]. We note that each Pauli gadget has one arbitrary angle rotation gate ($RZ(\theta)$), and thus the entire circuit has $\sim l/\epsilon$ such rotation gates. Here, ϵ is the error due to inadequate representation of eigenvalues of the Hamiltonian via limited n_r , and goes as $\sim 2^{-n_r}$. For an admissible error of δ in transpiling an arbitrary angle Z rotation into one-qubit Cliffords and T gates, one requires $\sim \log(1/\delta)$ T gates. Thus, the T count for QPE-CASCI is $\sim \frac{l}{\epsilon} \log(\frac{1}{\delta})$.

We now turn our attention to resource estimates from QPE-CASCI, presented in Fig. 9(c), where we observe trends similar to those with the VQE algorithm, where the requirement on the number of logical qubits and the number of CX gates are fewer for MSTO bases than for the 6-31G (about 70% of the qubits and about 6% of the CX gates) or cc-pVDZ (about 50% of the qubits and

about 0.25% of the CX gates) bases (also see supporting data in Table XI of Appendix C).

C. HHL-LCCSD

Finally, we take a look at the CX gate counts and energies from the HHL-LCCSD algorithm. The algorithm itself requires two QPE modules and a controlled-rotation module (which we shall call CR for brevity), but with amplitude encoding, the number of qubits in the state register is only $\lceil \log_2(n_s^4) \rceil$, as opposed to n_s in the case of QPE. Thus, the number of CX gates from one QPE module comes to $\approx 2^{n_r} 4l \lceil \log_2(n_s^4) \rceil + n_r^2/2 + n_r$. We now move to the CX estimates for the CR module. The circuit has 2^{n_r} multi-controlled RY gates, each with n_r control qubits, and thus it has a decomposition involving $\approx 2^{n_r+2} - 4n_r - 4$ CX gates [44]. Hence, the net gate count from HHL-LCCSD comes out to be $\approx 2^{2n_r+4} l \lceil \log_2(n_s^4) \rceil + n_r^2/2 + n_r + (2^{n_r+2} - 4n_r - 4)$. Once again, we recall that n_r only represents the precision in representing the eigenvalues of A to n_r -bit precision, and thus does not scale with system size. Since l goes as n_s^4 , the HHL-LCCSD circuit has number of CX gates that are quartic in the number of spin orbitals.

We add a few remarks:

- Our estimates exclude the cost of preparing a unitary V whose action on $|0\rangle^{\lceil \log_2(n_s^4) \rceil}$ yields $|b\rangle$. The possibility of efficient input state preparation is an open problem in QPE and algorithms that build on it. It could also occur in VQE when we employ multi-reference ansätze (for example, see Refs. [17, 45]).
- We also exclude the cost of extracting a molecular property as a feature of the solution ket, $|x\rangle \approx A^{-1}|b\rangle$ that HHL prepares. Let us consider the example of ground state energy calculation using the CSWAP test, whose inputs are $|b\rangle$ and $|x\rangle$ (output of HHL). The circuit requires $\lceil \log_2(n_s^4) \rceil$ CSWAP gates, and since each CSWAP is known to have a straightforward decomposition into 8 CX gates, the net CX gate count from the feature extraction module for obtaining ground state energies is $\approx 8 \lceil \log_2(n_s^4) \rceil$.
- We expect the T gate count to scale as that for QPE, that is, $\sim \frac{1}{\epsilon} \log(\frac{1}{\delta})$. This is because the CR module is cheaper; the circuit is expected to have $\sim 2^{n_r}$ rotation gates in its decomposition, each with $\sim \log(\frac{1}{\delta})$ T gates, and thus using $2^{n_r} = \epsilon$, we obtain $\sim \frac{1}{\epsilon} \log(\frac{1}{\delta})$.

We now move to numerical estimates of CX counts and ground state energies. We find from Fig. 9(d) that the results are very similar in terms of trends observed for QPE (see supporting data in Table XII of Appendix C), indicating that MSTO basis sets are an excellent choice for

obtaining good quality energies with much fewer quantum resources.

VI. COMPARISON WITH EARLIER WORKS IN LITERATURE

Now that we have discussed our results, it is in order to survey the existing related works in literature, and compare our work in relation to them. We note that earlier efforts towards refining and generating better single particle basis sets were not typically aimed at quantum computing applications, with the exception of one work.

We begin with the very first work that introduced the minimal basis sets [27], and in particular, the basis sets STO-3G through STO-6G for H through F with the exception of He. They use the method of least squares to obtain the exponents and contraction coefficients. We find that our MSTO- K G results are consistently lower than their STO- K G and their STO (the actual Slater type orbitals) results. It is also important to note that the STO- K G results from Ref. [27] and those obtained using the basis sets from Basis Set Exchange database differ slightly, with the latter being lower than the former. In our data, when we make a comparison between STO- K G and MSTO- K G bases, we use the refined STO- K G values provided in the Basis Set Exchange database.

The work by Tavouksoglou and Huzinaga [46] improve upon the STO-3G basis by using the method of least squares. We find that our MSTO-3G results are lower than those obtained in this work for all of the atoms that they consider: Li through F.

We now discuss a work [34] that is partially similar in spirit to ours, where the authors optimize STO-3G and STO-6G basis sets for the first row elements of the periodic table, using a generalized simulated annealing based approach. They optimize the Hartree-Fock functional, and the parameters that are optimized are not only the exponents and contraction coefficients but also the coefficients of atomic orbitals in the linear combination of atomic orbitals (LCAO) approach. We deviate in that we use a genetic algorithm with aggressive optimization by beginning with existing STO series bases, and do so for STO-3G through 6G and extend it up to a cardinal number of 11. We find that our STO-3G and 6G and their STO-3G and 6G HF energies match exactly for all atoms, serving as a sanity check that our STO calculations are correct. At the STO-3G level, all of our MSTO HF energies are lower than theirs, except for the case of H, where our energies exactly match. Especially for larger systems considered, we obtain lower energies at ~ 0.1 mHa level. At the STO-6G level, all of our MSTO energies are once again lower than theirs, except for H, where they obtain slightly better quality (lower) energy than ours (differing at ~ 0.1 mHa level). We also note that we extend our basis set generation beyond $k=6$, up to 11. Fig. 10 compares the energies obtained for atoms Li through F using our MSTO-3G and MSTO-6G bases

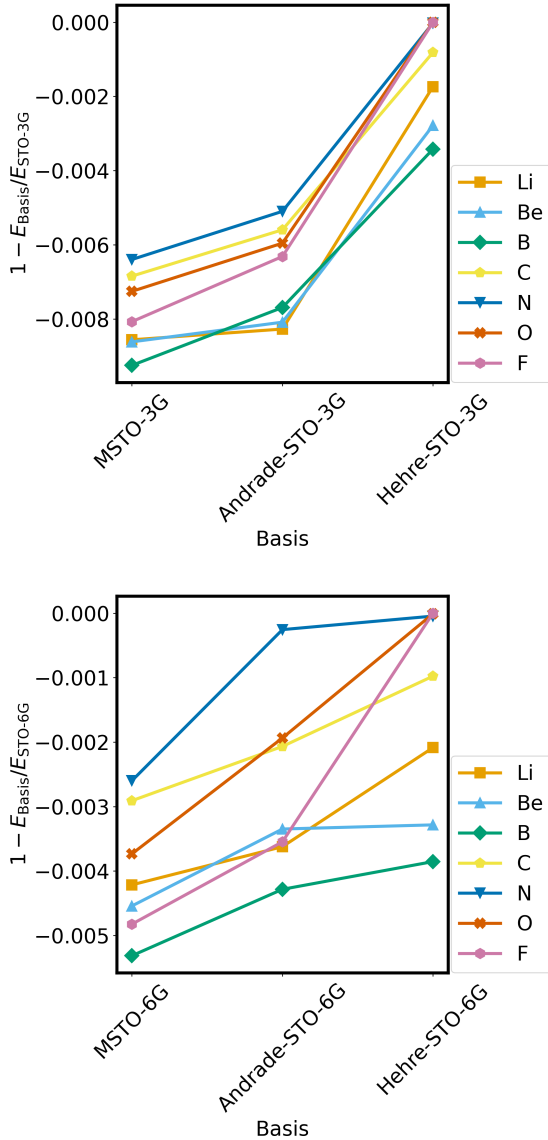


FIG. 10. Change in ground state energies for Li through F using our basis, as well as those from Refs. [27] and [34] for cardinal numbers $k = 3$ (top) and 6 (bottom) with respect to the STO- k G basis set.

with results from their 3G and 6G quality bases in Refs. [27] and [34], as a visual indicator of the performance of our basis sets relative to these two works.

In Ref. [26], the author uses both least squares fit and energy minimization approaches for the H atom for k of 2, 3, and 5, and finds that the latter approach yields lower energies. The results from this work are consistent with those obtained using MSTO- k G basis for $k=2, 3$, and 5. However, the work is limited to only the H atom and up to $k=5$.

Kapusta *et al* [47] have refined the existing STO-3G bases and generated their STO(0)-3G basis sets for principal quantum numbers from 1 to 7. They employ a

minimized sum of the squared deviations approach to carry out their optimization. We note that a limitation of this approach is assigning the same exponents and contraction coefficients for any atom for a given orbital. We compare our HF MSTO-3G results and theirs using STO(0)-3G for the atoms that we consider, namely H, C, N, O, and F, and find that our ground state energies are consistently lower than theirs. In particular, we note that since an important focus of their work was getting better energies for the atoms C, N, O, and H as they are the main building blocks of several organic molecules, one could employ our basis sets for large quantum chemical calculations involving molecules built out of these atoms. Having mentioned that, recalling the limitation in capturing correlation effects for minimal type bases, we recommend employing the more expensive 6-31G basis sets for such computations whenever feasible computationally. Reoptimizing the 6-31G bases with our algorithm is a possibility, but is outside the scope of the current study. Preliminary investigations in that direction indicate that the 6-31G bases are already well-optimized, and improvements are at most $\sim O(1)$ milliHartree.

We finally turn our attention to a relatively recent work from 2023 [48], where the authors optimize the exponents and contraction coefficients of the STO-3G basis for every bond distance to obtain potential energy surfaces of the hydrogen molecule with the goal of executing VQE on an IBM superconducting qubit quantum computer. They restrict themselves to only the hydrogen molecule and $K=3$ in their study. We do not carry out an adaptive calculation in our work but is very much possible in principle in future studies. On this note, we also add that a very interesting extension of our work could be in the direction of developing modified mixed ramp Gaussian basis sets [49, 50].

VII. CONCLUSION

In summary, we have designed our memetic algorithm (a genetic algorithm followed by aggressive refinement) to construct modified minimal basis sets with cardinal numbers 2 through 11 (MSTO- k G; $k = 2$ through 11), for atoms H through F, excluding He. The MSTO basis sets are intended for quantum computing applications, where we are expected to be limited by qubit quality in the coming few years. Our work builds on the expectation that sufficiently reoptimized minimal basis sets between k of 3 and 6 should yield better (lower) energies than their existing versions, and further exploits the fact that in their contracted form, minimal basis sets with increasing cardinal numbers retain the same number of spin orbitals for a given atom, and therefore the same qubit footprint in a quantum computing calculation, while yielding better (lower) ground state energies.

We find that our MSTO- k G basis sets consistently outperform the minimal basis sets and deliver FCI ground state energies comparable to or lower than those obtained

with the 6-31G basis for the considered atomic systems. This gain comes with fewer spin orbitals, 10 in our sets versus 18 in the 6-31G basis. Consequently, not only do we expect our MSTO- k G basis sets to perform comparable to or outperform split-valence basis sets in quantum chemistry applications on quantum computers, but also in large molecular systems in conventional quantum chemical computations with far fewer computational resources.

Of particular interest are our results for Li and Be; the ground state energies are lower than those obtained using Dunning basis sets. Specifically, the result for Li surpasses even the cc-pVQZ result, and resolves the correlation energy to about half way relative to state-of-the-art benchmark few-body calculations. On the other hand, the MSTO bases inherit the limitation of not having sufficient spin orbitals for the elements to the right of the periodic table (N, O, and F); although the base HF energies are better than their STO and 6-31G counterparts, the FCI energies from MSTO bases are near zero and thus are higher than 6-31G FCI energies.

We also compute molecular potential energy curves for H_2 , Li_2 , C_2 , LiH , BeH and BeH_2 molecules, and find that while the results for H_2 are poor (an observation consistent with an earlier work [34]), the results for other molecules are at least comparable to those from 6-31G bases.

We carry out a simple resource estimation exercise and compare the performance of the MSTO- k G basis and the 6-31G basis in terms of number of CX gates versus the predicted ground state energies with the VQE-UCCSD, QPE-CASCI, and the HHL-LCCSD algorithms for Li taken as a best-case example. Since the number of CX gates per expectation value evaluation per iteration for VQE-UCCSD and also the gate count for QPE-CASCI scale as $\sim n_s^5$ in n_s number of spin orbitals, the CX gate count dependence on basis set change is quite significant, and thus obtaining good quality energies with far fewer CX gates makes a notable difference in practical noisy quantum computing calculations in the coming decade or so. With HHL-LCCSD, the CX gate count scales as $\sim n_s^4[\log(n_s)]$, and while it is not as sensitive as the

earlier two algorithms, here too, choice of single particle basis sets impact the gate counts noticeably. We add that the T counts scale as $\sim \frac{n_s^4}{\epsilon} \log(\frac{1}{\delta})$ for QPE-CASCI and HHL-LCCSD, thus resulting in a quartic dependence of the number of spin orbitals, that is, the number of basis elements, for the T count. Thus, the choice of basis set significantly (polynomial in spin orbitals) impacts the gate counts for VQE, QPE, and HHL applied to chemistry.

We then compare our methods and bases with related works in literature that generate modified minimal basis sets, and find that our basis sets outperform all of them in terms of predicting atomic and molecular ground state energies.

Given the usefulness of classical pre-processing to reduce quantum resources, and combined with the one-time nature of generating useful basis sets for this purpose with state-of-the-art optimization algorithms, we plan to extend in future works our basis sets to larger atomic systems with newer and more high performance classical optimization methods. With improved algorithms, we also plan to reoptimize some of the larger basis sets available in literature.

DATA AVAILABILITY

The codes that we developed are open-source, and can be accessed here: <https://github.com/subimal/MSTO-kG>.

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Appendix A: Computed MSTO- k G basis sets

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
H	0	2	1.332500	0.335072	H	0	7	36.215701	0.004636	H	0	10	0.023408	0.018613
			0.201530	1.002799				5.945957	0.029019				47.451794	0.011323
H	0	3	4.492911	0.062664				1.672836	0.086907				8.632545	0.060140
			0.680729	0.362123				0.372023	0.302611				0.266933	1.278035
			0.151340	0.574683				0.145877	0.346923				0.080237	0.026456
H	0	4	12.433109	0.013700				0.058664	0.083617				0.070984	0.248660
			1.903457	0.093517				0.794523	0.107805				0.495950	0.089100
			0.436273	0.317705	H	0	8	57.360124	0.010589				0.787069	0.674741
			0.120787	0.326516				8.858612	0.075640				0.114002	0.727133
H	0	5	24.561782	0.015833				0.573397	0.145787				2.425374	0.230988
			3.924283	0.105252				0.697915	0.620006	H	0	11	8.956817	0.359687
			0.957988	0.416065				0.241767	1.575516				52.454669	0.058020
			0.286151	0.905476				0.705874	0.139903				1.525339	0.450066
			0.094807	0.590050				0.088118	0.938748				0.066580	1.969280
H	0	6	42.431848	0.005984				2.154795	0.319647				0.063431	0.514916
			6.527242	0.043765	H	0	9	66.397548	0.008338				0.014178	-0.013964
			1.527203	0.194622				10.149185	0.061116				0.301358	1.933829
			0.448575	0.523163				0.249153	0.235949				2.301479	1.354177
			0.153500	0.610899				0.794520	0.448464				0.638397	4.687667
			0.057351	0.131116				0.225253	1.107420				0.297515	1.404231
								0.806379	0.235587				0.166707	6.730856
								0.087305	0.867243					
								2.483953	0.259263					
								0.413336	0.419480					

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
Li	0	2	10.443437	0.329496	Li	0	7	343.220752	0.005357	Li	0	10	458.892700	0.010790
			1.438243	1.009065				54.985322	0.036420				77.955495	0.065325
Li	0	3	34.979819	0.077973				13.360790	0.148893				20.983594	0.239748
			5.191709	0.420472				3.986245	0.383807				7.359284	0.557524
			1.059549	0.702156				0.774205	0.155673				3.126922	0.672852
Li	0	4	97.695864	0.032939				0.414568	0.157633				0.413084	0.192311
			14.820762	0.218052				1.394985	0.478386				1.336112	1.306319
			3.266465	0.701671	Li	0	8	421.892873	0.009975				0.095085	0.026357
			0.834229	0.786918				65.887804	0.069748				0.584830	0.619891
Li	0	5	232.706805	0.009798				15.832399	0.294670				3.052762	0.269462
			36.039372	0.068770				3.211819	0.040798	Li	0	11	333.146237	0.014048
			8.326178	0.279340				4.929907	0.667753				5.119031	0.341142
			2.293087	0.637974				0.450908	0.504495				2.163372	0.420482
			0.686717	0.523345				2.564102	0.429413				22.553178	0.146828
Li	0	6	357.319648	0.005192				1.233569	1.204849				0.020839	1.097782
			56.415381	0.035738	Li	0	9	0.011363	0.256536				0.024182	0.015530
			13.458931	0.151421				0.641313	-1.525312				0.027150	0.032018
			3.901728	0.404018				15.993439	0.057106				0.020127	0.185951
			1.262537	0.565961				0.029647	7.404341				0.154551	0.133746
			0.443288	0.210397				0.029612	0.393366				0.052073	5.182478
								0.028809	0.448355				0.116862	0.334861
								0.072790	9.374863					
								175.847312	0.005491					
								0.011996	0.216768					

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
Be	0	2	19.490692	0.180891	Be	0	7	656.902472	0.006197	Be	0	10	1008.657710	0.007988
			2.750151	0.563571				104.195338	0.042957				163.024724	0.053555
Be	0	3	65.687087	0.102922				24.749232	0.185009				4.185682	1.325653
			9.803806	0.569129				7.175726	0.503304				40.371418	0.231626
			2.042436	0.958605				2.331965	0.696914				0.070970	0.007655
Be	0	4	187.952501	0.028393				0.808728	0.234208				0.874405	0.005308
			28.348333	0.192973				0.265758	-0.004638				0.442956	0.231571
			6.280651	0.642124	Be	0	8	1.886943	0.271926				0.083053	0.232829
			1.624882	0.732551				617.363676	0.002630				12.227800	0.694232
Be	0	5	434.091838	0.009155				5.994324	0.251085	Be	0	11	1.527315	1.166832
			67.177289	0.064447				5.359786	-0.025928				866.869580	0.009967
			15.639776	0.263853				93.888423	0.019227				139.190385	0.067661
			4.357526	0.615606				0.541312	0.048853				3.555907	0.678993
			1.325476	0.501686				21.626925	0.084649				34.162684	0.288293
Be	0	6	628.105510	0.005689				0.131021	-0.019203				0.069881	0.006790
			99.611392	0.038544	Be	0	9	611.309439	0.008527				1.231278	0.009922
			24.211405	0.160504				0.624954	0.161003				0.254566	0.223872
			7.160029	0.428508				94.986364	0.060176				0.068231	0.223153
			2.362672	0.590722				3.229133	0.120667				10.366518	0.825035
			0.848907	0.211013				1.900250	0.798600				1.287208	1.044861
								0.090549	0.015221				3.563254	0.762835
								22.357310	0.259372					
								0.507804	0.010308					
								6.444248	0.663903					

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
B	0	2	31.391771	0.362265	B	0	7	895.059309	0.006588	B	0	10	1220.127360	0.016194
			4.497927	1.138905				11.051797	0.446943				200.766320	0.104950
B	0	3	105.903536	0.080683				146.019232	0.042622				50.341058	0.438146
			15.835665	0.452640				36.492356	0.171727				5.308831	2.032173
			3.340707	0.764567				3.736563	0.596800				15.368965	1.250279
B	0	4	303.839820	0.023912				1.245423	0.078461				0.042953	-0.014611
			45.839141	0.163899				1.433852	0.127030				0.094774	0.815196
			10.211701	0.552840	B	0	8	979.220174	0.007804				1.898206	1.259487
			2.669101	0.634086				150.870702	0.055934				1.872614	0.034049
B	0	5	708.921534	0.008755				35.319511	0.242085	B	0	11	0.308397	1.030609
			109.223926	0.062082				10.046125	0.635492				1328.596970	0.020229
			25.326119	0.259488				2.867956	0.568970				209.008919	0.140702
			7.067193	0.612300				3.999577	0.201235				49.886708	0.628299
			2.166124	0.494621				0.123791	0.628966				4.896350	2.834152
B	0	6	1191.995380	0.003906				0.482255	0.430230				14.563393	1.849380
			187.930922	0.027353	B	0	9	775.940317	0.007908				0.024138	-0.009807
			44.719153	0.120466				124.730600	0.052383				0.298040	0.770627
			12.983995	0.345787				30.452505	0.211618				2.102849	0.948921
			4.260820	0.519448				9.027329	0.519359				2.631211	0.042461
			1.515436	0.220620				2.881014	0.484591				0.103582	0.766941
								0.307805	0.162751				1.292456	0.727593
								0.115726	0.803441					
								1.023049	0.036758					
								0.406167	0.446557					

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
C	0	2	46.098325	0.191217	C	0	7	1194.188930	0.006589	C	0	10	1582.633300	0.009592
			6.670982	0.604580				194.061751	0.042889				257.921838	0.063061
C	0	3	155.296401	0.063022				48.028224	0.171972				64.302206	0.260295
			23.288858	0.356559				14.522873	0.431488				19.811809	0.705482
			4.947649	0.602982				4.975868	0.514307				7.061145	1.021587
C	0	4	447.313496	0.026621				1.910933	0.135474				2.358115	0.014973
			67.341312	0.183823				1.293437	0.011825				2.051796	0.162563
			15.018155	0.627207	C	0	8	1935.542100	0.009644				0.559670	0.325054
			3.946006	0.714807				308.793054	0.067021				0.171192	0.392110
C	0	5	1007.888150	0.009996				73.034477	0.307667				2.913736	0.428219
			156.503898	0.070181				20.884887	0.942475	C	0	11	1864.440580	0.004698
			36.465519	0.292900				6.695430	1.553847				299.345959	0.031956
			10.261023	0.685502				0.051813	0.000818				72.890615	0.138828
			3.174388	0.541710				1.947105	0.719484				21.873407	0.398051
C	0	6	1487.729180	0.005473				0.238017	0.524677				2.686373	0.369809
			237.875623	0.037129	C	0	9	1066.069100	0.011969				2.088374	0.004781
			57.569062	0.158239				166.093230	0.083659				6.826172	0.423071
			16.957780	0.434670				2.052613	0.050999				0.120044	0.059999
			5.638156	0.596224				11.279091	0.824023				0.555554	0.105775
			2.016810	0.204718				39.120476	0.347334				8.567850	0.240747
								3.595335	0.319724				0.246686	0.081753
								1.807489	0.036742					
								0.174701	-0.014630					
								3.784566	0.386289					

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
N	0	2	63.565489	0.309760	N	0	7	1751.597230	0.005324	N	0	10	1707.091920	0.007121
			9.263207	0.982451				272.872768	0.037455				279.323662	0.046235
N	0	3	213.361835	0.081482				64.619601	0.159329				68.288578	0.175184
			32.067919	0.462794				18.897945	0.401234				21.962916	0.446652
			6.854283	0.780701				6.763559	0.354623				7.237614	0.377013
N	0	4	614.349623	0.025689				4.345101	0.137402				2.692827	0.150678
			92.690354	0.177481				1.721191	0.058985				0.308686	0.661927
			20.760638	0.607633	N	0	8	1878.683780	0.005811				0.595719	0.608329
			5.479167	0.692917				297.849168	0.039956				0.782707	1.108288
N	0	5	1245.737030	0.010293				71.805053	0.168733				0.194694	1.219721
			195.678174	0.070354				21.297556	0.446982	N	0	11	2116.699360	0.013453
			46.371297	0.283852				7.122089	0.557259				344.280780	0.087776
			13.334402	0.628193				2.701200	0.155555				89.281609	0.324661
			4.232987	0.461623				0.652792	0.027782				32.653984	0.644270
N	0	6	1878.144370	0.005565				0.185818	0.003580				14.167867	1.136206
			300.246020	0.037801	N	0	9	1736.915230	0.007053				2.891806	0.016779
			72.678832	0.159332				284.402618	0.045798				0.382982	1.383056
			21.469328	0.425158				71.344937	0.183229				5.329930	1.116807
			7.156883	0.542742				22.061050	0.462890				1.198003	0.558085
			2.519045	0.152024				8.020781	0.443273				0.129756	0.450012
								3.163662	0.241639				0.991897	0.179739
								0.199798	0.764996					
								0.714970	0.847771					
								0.421315	0.222484					

Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
O	0	2	83.878966	0.129142	O	0	7	2573.046670	0.006303	O	0	10	2371.244800	0.008166
			12.286765	0.410711				414.765540	0.042092				394.712258	0.051138
O	0	3	282.356352	0.088788				102.059410	0.175262				100.660616	0.203302
			42.456533	0.507544				30.883247	0.470288				31.045853	0.549988
			9.100558	0.859179				10.595279	0.639573				10.652512	0.507803
O	0	4	812.095834	0.026268				4.036538	0.237306				3.831247	0.289728
			122.440542	0.182070				1.795940	0.013175				0.146480	0.579222
			27.467276	0.626194	O	0	8	2050.376470	0.007187				1.142125	0.919614
			7.267088	0.714723				336.608199	0.046225				0.411196	1.011240
O	0	5	1622.509400	0.011464				84.270010	0.183324				0.626314	1.064588
			255.457870	0.078649				26.072692	0.443769	O	0	11	2177.017900	0.014633
			60.488003	0.317980				8.668950	0.449690				348.585098	0.098007
			17.429605	0.701919				3.126057	0.120443				88.269991	0.355623
			5.546655	0.503621				0.289788	0.662240				18.483027	0.935639
O	0	6	2459.678270	0.005909				0.960094	0.720026				36.670367	0.412957
			393.874043	0.039799	O	0	9	2294.065450	0.004467				6.562876	0.066680
			96.232398	0.166137				389.805404	0.027256				0.292368	1.230669
			28.753679	0.442812				101.694996	0.105132				1.239853	0.385162
			9.662396	0.574660				32.243704	0.275416				6.227805	0.817976
			3.427536	0.169654				11.694377	0.325448				0.994733	0.791850
								4.792696	0.170558				0.661935	0.276047
								0.242648	0.230140					
								0.541727	0.279910					
								1.200351	0.261935					
Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient	Element	l	k	Exponent	Coefficient
F	0	2	106.959844	0.207832	F	0	7	2924.910650	0.005546	F	0	10	2967.099950	0.006456
			15.729023	0.661991				473.005150	0.036708				474.871805	0.043540
F	0	3	359.860551	0.082573				116.797193	0.149967				115.705521	0.181503
			54.172371	0.473389				35.608673	0.386072				34.696268	0.472152
			11.647123	0.801526				12.302986	0.480870				11.240044	0.494282
F	0	4	1024.126410	0.026796				4.679761	0.151294				4.266532	0.128876
			154.966564	0.185088				1.401899	0.007168				0.313039	0.418206
			34.893196	0.635767	F	0	8	3386.120700	0.006014				0.704657	0.300246
			9.269533	0.721131				545.486408	0.040340				0.684285	0.051680
F	0	5	2171.499400	0.009929				134.173685	0.168956				1.379664	0.499433
			339.437935	0.068891				40.566453	0.463858	F	0	11	3251.709790	0.005580
			80.092451	0.283092				13.771119	0.628601				529.781291	0.036680
			23.017945	0.641438				4.740271	0.263775				132.184122	0.149956
			7.270549	0.481901				0.951284	0.232965				40.882158	0.397161
F	0	6	3101.056010	0.005660				0.287062	0.115580				14.878813	0.324266
			498.931583	0.037859	F	0	9	3167.615710	0.006829				5.767360	0.214250
			122.468326	0.157729				495.813326	0.047740				1.516486	0.701776
			36.651305	0.421556				117.937089	0.206538				0.912744	0.266157
			12.384611	0.543017				34.351807	0.559036				0.793492	0.462920
			4.432168	0.164265				11.131686	0.670817				0.340810	1.225594
								3.587322	0.147315				1.153033	0.467925
								1.510421	0.030387					
								0.477706	0.064078					
								0.014773	0.004474					

Appendix B: Comparison of basis set energies

TABLE I. Comparison of FCI energies (in Ha) for full active space $E(FCI)$ versus active space of 10 spin orbitals $E(asFCI)$. We employed the ccpV-DZ basis sets for these calculations.

Atom	E (HF)	E (asFCI)	E_{cor} (asFCI)	E(FCI)	E_{cor} (FCI)	% E_{cor} (lost)
B	-24.526591	-24.545417	-0.018826	-24.590630	-0.064039	70.601654
C	-37.682418	-37.694278	-0.011860	-37.761905	-0.079487	85.079009
N	-54.388414	-54.388414	0.000000	-54.480115	-0.091701	100.000000
O	-74.787513	-74.787513	0.000000	-74.911744	-0.124231	100.000000
F	-99.371862	-99.371862	0.000000	-99.529518	-0.157656	100.000000

TABLE II. Comparison of basis set energies (Ha) for H.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-0.454397	-0.454397	0.000000
3	STO	-0.466582	-0.466582	0.000000
4	STO	-0.469806	-0.469806	0.000000
5	STO	-0.470743	-0.470743	0.000000
6	STO	-0.471039	-0.471039	0.000000
2	MSTO	-0.485813	-0.485813	0.000000
3	MSTO	-0.496979	-0.496979	0.000000
4	MSTO	-0.499277	-0.499277	0.000000
5	MSTO	-0.499788	-0.499788	0.000000
6	MSTO	-0.499897	-0.499897	0.000000
7	MSTO	-0.499899	-0.499899	0.000000
8	MSTO	-0.499934	-0.499934	0.000000
9	MSTO	-0.499945	-0.499945	0.000000
10	MSTO	-0.499936	-0.499936	0.000000
11	MSTO	-0.499934	-0.499934	0.000000
	6-31G	-0.498233	-0.498233	0.000000
	cc-pVDZ	-0.499278	-0.499278	0.000000
	cc-pVTZ	-0.498233	-0.498233	0.000000
	cc-pVQZ	-0.498233	-0.498233	0.000000
3	Andrade-STO	-0.496979		
6	Andrade-STO	-0.499931		
3	Hehre-STO	-0.494910		
4	Hehre-STO	-0.498480		
5	Hehre-STO	-0.499510		
6	Hehre-STO	-0.499830		

TABLE III. Comparison of basis set energies (Ha) for Li. The Hylleraas result is not quite FCI, but is placed under that column for convenience.

k	Basis	HF	FCI	E_{cor} (FCI)
	STO	-7.070821	-7.071101	-0.000280
3	STO	-7.315526	-7.315837	-0.000311
4	STO	-7.376840	-7.377151	-0.000311
5	STO	-7.394279	-7.394588	-0.000309
6	STO	-7.399931	-7.400238	-0.000307
2	MSTO	-7.180509	-7.203128	-0.022619
3	MSTO	-7.378140	-7.399996	-0.021856
4	MSTO	-7.419284	-7.441139	-0.021855
5	MSTO	-7.429019	-7.450625	-0.021606
6	MSTO	-7.431114	-7.452950	-0.021836
7	MSTO	-7.431154	-7.452979	-0.021825
8	MSTO	-7.431452	-7.453074	-0.021622
9	MSTO	-7.431357	-7.453205	-0.021847
10	MSTO	-7.431858	-7.453678	-0.021820
11	MSTO	-7.430544	-7.452345	-0.021801
	6-31G	-7.431235	-7.431554	-0.000319
	cc-pVDZ	-7.432420	-7.432638	-0.000218
	cc-pVTZ	-7.431235	-7.431554	-0.000319
	cc-pVQZ	-7.431235	-7.431554	-0.000319
3	Andrade-STO	-7.376022		
6	Andrade-STO	-7.426732		
3	Hehre-STO	-7.328230		
4	Hehre-STO	-7.391850		
5	Hehre-STO	-7.409710		
6	Hehre-STO	-7.415360		
	STO-exact	-7.418480		
	Hylleraas ∞		-7.4780603239	

TABLE IV. Comparison of basis set energies for Be.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-13.890237	-13.943417	-0.053180
3	STO	-14.351880	-14.403655	-0.051775
4	STO	-14.462363	-14.514842	-0.052479
5	STO	-14.493169	-14.545888	-0.052719
6	STO	-14.503361	-14.556089	-0.052727
2	MSTO	-14.116338	-14.160902	-0.044564
3	MSTO	-14.475514	-14.519908	-0.044394
4	MSTO	-14.548667	-14.593126	-0.044458
5	MSTO	-14.565866	-14.610372	-0.044506
6	MSTO	-14.569223	-14.613735	-0.044512
7	MSTO	-14.569395	-14.613934	-0.044540
8	MSTO	-14.568066	-14.612231	-0.044165
9	MSTO	-14.568849	-14.613135	-0.044286
10	MSTO	-14.570482	-14.614123	-0.043641
11	MSTO	-14.570215	-14.614487	-0.044273
	6-31G	-14.566764	-14.613545	-0.046781
	cc-pVDZ	-14.572338	-14.617410	-0.045072
	cc-pVTZ	-14.566764	-14.613545	-0.046781
	cc-pVQZ	-14.566764	-14.613545	-0.046781
3	Andrade-STO	-14.467908		
6	Andrade-STO	-14.551913		
3	Hehre-STO	-14.391800		
4	Hehre-STO	-14.508840		
5	Hehre-STO	-14.540800		
6	Hehre-STO	-14.550980		
	STO-exact	-14.556700		

TABLE V. Comparison of basis set energies (Ha) for B.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-23.395284	-23.436414	-0.041130
3	STO	-24.148989	-24.189265	-0.040276
4	STO	-24.329130	-24.369947	-0.040818
5	STO	-24.378370	-24.419391	-0.041022
6	STO	-24.394295	-24.435336	-0.041042
2	MSTO	-23.792264	-23.824089	-0.031826
3	MSTO	-24.372327	-24.404165	-0.031838
4	MSTO	-24.490291	-24.522259	-0.031968
5	MSTO	-24.518191	-24.550106	-0.031915
6	MSTO	-24.523900	-24.555776	-0.031876
7	MSTO	-24.522879	-24.554883	-0.032004
8	MSTO	-24.521440	-24.553409	-0.031968
9	MSTO	-24.520364	-24.552349	-0.031985
10	MSTO	-24.524002	-24.555875	-0.031873
11	MSTO	-24.524135	-24.556048	-0.031913
	6-31G	-24.519348	-24.562892	-0.043544
	cc-pVDZ	-24.526591	-24.590630	-0.064039
	cc-pVTZ	-24.519348	-24.562892	-0.043544
	cc-pVQZ	-24.519348	-24.562892	-0.043544
3	Andrade-STO	-24.334665		
6	Andrade-STO	-24.498767		
3	Hehre-STO	-24.231600		
4	Hehre-STO	-24.422160		
5	Hehre-STO	-24.472260		
6	Hehre-STO	-24.488290		
	STO-exact	-24.497170		

TABLE VI. Comparison of basis set energies (Ha) for C.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-36.060274	-36.080845	-0.020571
3	STO	-37.198393	-37.218734	-0.020341
4	STO	-37.475329	-37.495739	-0.020410
5	STO	-37.549193	-37.569674	-0.020481
6	STO	-37.572364	-37.592856	-0.020492
2	MSTO	-36.578039	-36.595428	-0.017389
3	MSTO	-37.452952	-37.470378	-0.017426
4	MSTO	-37.631161	-37.648613	-0.017452
5	MSTO	-37.672867	-37.690315	-0.017448
6	MSTO	-37.681636	-37.699097	-0.017461
7	MSTO	-37.679403	-37.696870	-0.017466
8	MSTO	-37.676932	-37.694540	-0.017608
9	MSTO	-37.673972	-37.691481	-0.017510
10	MSTO	-37.681713	-37.699199	-0.017485
11	MSTO	-37.682486	-37.699960	-0.017474
	6-31G	-37.676866	-37.716264	-0.039399
	cc-pVDZ	-37.682418	-37.761905	-0.079487
	cc-pVTZ	-37.676866	-37.716264	-0.039399
	cc-pVQZ	-37.676866	-37.716264	-0.039399
3	Andrade-STO	-37.406499		
6	Andrade-STO	-37.649999		
3	Hehre-STO	-37.228660		
4	Hehre-STO	-37.510690		
5	Hehre-STO	-37.585780		
6	Hehre-STO	-37.609060		
	STO-exact	-37.621820		

TABLE VII. Comparison of basis set energies (Ha) for N.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-52.097007	-52.097007	0.000000
3	STO	-53.719010	-53.719010	0.000000
4	STO	-54.113775	-54.113775	0.000000
5	STO	-54.217310	-54.217310	0.000000
6	STO	-54.249112	-54.249112	0.000000
2	MSTO	-52.808559	-52.808559	0.000000
3	MSTO	-54.062540	-54.062540	0.000000
4	MSTO	-54.318772	-54.318772	0.000000
5	MSTO	-54.377674	-54.377674	0.000000
6	MSTO	-54.390099	-54.390099	0.000000
7	MSTO	-54.388022	-54.388022	0.000000
8	MSTO	-54.386761	-54.386761	0.000000
9	MSTO	-54.387388	-54.387388	0.000000
10	MSTO	-54.386063	-54.386063	0.000000
11	MSTO	-54.389492	-54.389492	0.000000
	6-31G	-54.382051	-54.419940	-0.037889
	cc-pVDZ	-54.388414	-54.480115	-0.091701
	cc-pVTZ	-54.382051	-54.419940	-0.037889
	cc-pVQZ	-54.382051	-54.419940	-0.037889
3	Andrade-STO	-53.992810		
6	Andrade-STO	-54.262930		
3	Hehre-STO	-53.720100		
4	Hehre-STO	-54.115850		
5	Hehre-STO	-54.219720		
6	Hehre-STO	-54.251550		
	STO-exact	-54.268860		

TABLE VIII. Comparison of basis set energies (Ha) for O.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-71.572305	-71.572305	0.000000
3	STO	-73.804150	-73.804150	0.000000
4	STO	-74.336874	-74.336874	0.000000
5	STO	-74.474901	-74.474901	0.000000
6	STO	-74.516816	-74.516816	0.000000
2	MSTO	-72.602781	-72.602781	0.000000
3	MSTO	-74.339476	-74.339476	0.000000
4	MSTO	-74.695446	-74.695446	0.000000
5	MSTO	-74.776324	-74.776324	0.000000
6	MSTO	-74.795012	-74.795012	0.000000
7	MSTO	-74.795757	-74.795757	0.000000
8	MSTO	-74.790826	-74.790826	0.000000
9	MSTO	-74.794001	-74.794001	0.000000
10	MSTO	-74.792051	-74.792051	0.000000
11	MSTO	-74.789501	-74.789501	0.000000
	6-31G	-74.778234	-74.839408	-0.061174
	cc-pVDZ	-74.787513	-74.911744	-0.124231
	cc-pVTZ	-74.778234	-74.839408	-0.061174
	cc-pVQZ	-74.778234	-74.839408	-0.061174
3	Andrade-STO	-74.243618		
6	Andrade-STO	-74.661032		
3	Hehre-STO	-73.804250		
4	Hehre-STO	-74.337400		
5	Hehre-STO	-74.475550		
6	Hehre-STO	-74.517490		
	STO-exact	-74.540110		

TABLE IX. Comparison of basis set energies (Ha) for F.

k	Basis	HF	FCI	E_{cor} (FCI)
2	STO	-95.015084	-95.015084	0.000000
3	STO	-97.986505	-97.986505	0.000000
4	STO	-98.681687	-98.681687	0.000000
5	STO	-98.859721	-98.859721	0.000000
6	STO	-98.913253	-98.913253	0.000000
2	MSTO	-96.449071	-96.449071	0.000000
3	MSTO	-98.777281	-98.777281	0.000000
4	MSTO	-99.256406	-99.256406	0.000000
5	MSTO	-99.366828	-99.366828	0.000000
6	MSTO	-99.390320	-99.390320	0.000000
7	MSTO	-99.388497	-99.388497	0.000000
8	MSTO	-99.390137	-99.390137	0.000000
9	MSTO	-99.386926	-99.386926	0.000000
10	MSTO	-99.388069	-99.388069	0.000000
11	MSTO	-99.391395	-99.391395	0.000000
	6-31G	-99.360218	-99.447423	-0.087205
	cc-pVDZ	-99.371862	-99.529518	-0.157656
	cc-pVTZ	-99.360218	-99.447423	-0.087205
	cc-pVQZ	-99.360218	-99.447423	-0.087205
3	Andrade-STO	-98.605185		
6	Andrade-STO	-99.264015		
3	Hehre-STO	-97.987090		
4	Hehre-STO	-98.681850		
5	Hehre-STO	-98.859760		
6	Hehre-STO	-98.913270		
	STO-exact	-98.941930		

Appendix C: Comparison of quantum resources across basis sets

TABLE X. Resource requirements for different basis sets in a VQE-UCCSD calculation of finding Li atom's ground state energy. We also note that although we have provided $k = 6$ as a representative example, the quantum resources do not change across k for STO and for MSTO bases. We also provide FCI energy in place of the energy obtained using the quantum algorithm, for simplicity.

Resource	STO-6G	MSTO-11G	6-31G	cc-pVDZ
Number of spin orbitals	10	10	18	28
Number of qubits	10	10	18	28
Variational parameters	37	37	155	415
Number of CX gates	2246	2246	17,420	72,185
FCI energy (Ha)	-7.400238	-7.452345	-7.431554	-7.432638

TABLE XI. Qubit counts and CX gate estimates for different basis sets in a QPE-CASCI calculation of the ground state energy of the Li atom. The number of QPE clock register qubits is set to 8. JWT stands for Jordan-Wigner transformation.

Resource	STO-6G	MSTO-11G	6-31G	cc-pVDZ
Number of spin orbitals	10	10	18	28
Number of state register qubits (JWT)	10	10	18	28
Total number of qubits	18	18	26	36
Pauli terms	156	156	1492	22167
Number of CX gates	1,597,480	1,597,480	27,500,584	635,572,264
FCI energy (Ha)	-7.400238	-7.452345	-7.431554	-7.432638

TABLE XII. Qubit counts and CX gate estimates for different basis sets in a HHL-LCCSD calculation of the ground state energy of the Li atom. The number of QPE clock register qubits is set to 8.

Resource	STO-6G	MSTO-11G	6-31G	cc-pVDZ
Number of spin orbitals	10	10	18	28
Number of state register qubits	10	10	17	20
Total number of qubits	14	14	26	29
Pauli terms	156	156	1492	22167
Number of CX gates	4,473,860	4,473,860	51,946,500	907,961,348
FCI energy (Ha)	-7.400238	-7.452345	-7.431554	-7.432638

TABLE XIII. Symmetries and basis combinations that we studied (for di- and triatomic cases) of the molecules considered.

Molecule	State
H ₂	$1\Sigma^+$
Li ₂	$1\Sigma_g^+$
C ₂	$1\Sigma_g^+$
LiH	$1\Sigma^+$
BeH	$2\Sigma^+$
BeH ₂	$1\Sigma_g^+$

		Species B			
		STO-6G	MSTO-6G	MSTO-11G	6-31G
Species A	STO-6G	✓	✓	✓	✓
	MSTO-6G	✓	✓	✓	✓
	MSTO-11G	✓	✓	✓	✓
	6-31G	✓	✓	✓	✓